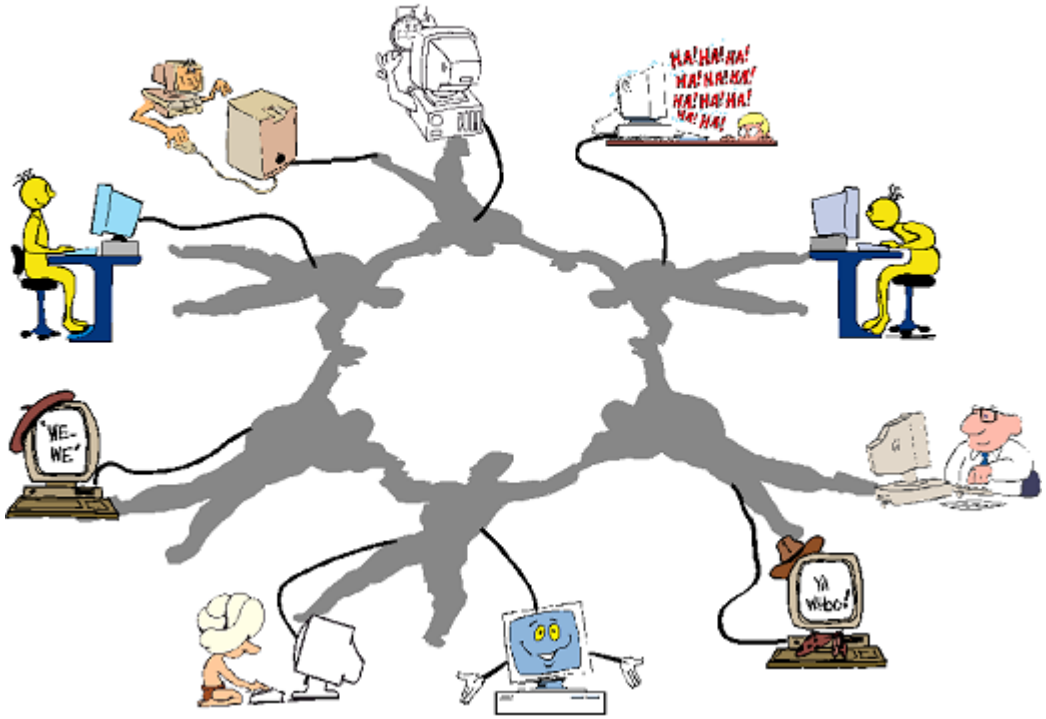
Computer Networks

Shmulik Kaplan

Slides by: Gadi Taubenfeld



Course information

Lectures:

Sunday 10:30 – 13:00 **English**

Monday 8:45 – 11:15; 12:45 – 15:15 **Hebrew**

Office Hours: Sunday 13:00 – 14:00 (immediately after the English class)

Lecturer: Shmulik Kaplan: Email: Shmuel.Kaplan01@post.runi.ac.il

TA: Daniel Karalnik

Course Resources:

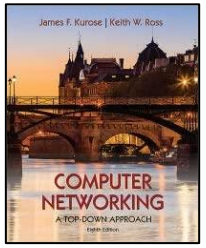
Moodle: Everything not in Piazza

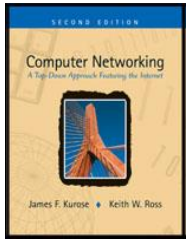
Piazza: Q&A forum, Announcements

Course information

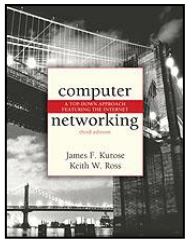
Textbook: Computer Networking
A Top-Down Approach Featuring the Internet, by Kurose & Ross

Additional written material: notes on distributed algorithms

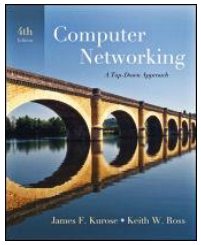




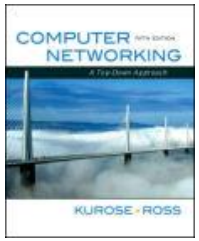
Second Edition



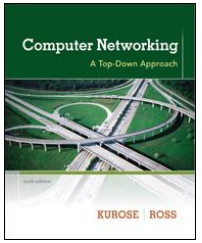
Third Edition



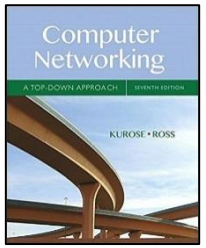
Forth Edition



Fifth Edition



Sixth Edition



Seventh Edition

The first two Lectures will cover Chapter 1 from the textbook

Homework & Grades



- General:
  - 5 exercises: 3X5=15%
  - Final exam: 85% (closed books)
  - Minimum final exam grade to pass 60
  - Strict late policies and policies on cheating
  - Bonus grade
- Ethics:
  - Collaboration in groups of 2-3 students is encouraged! But – take no written notes, no screenshots, no pictures, and write your own solution in your own words on your own.
- It is forbidden to use LLMs to produce any solutions to problems and exercises

Course Goals

- Understanding fundamental concepts of networking:
  - Design
  - Implementation
  - Protocols
  - Applications
- By the end of the course student should be able to:
  - Analyze a networking related problem.
  - Derive possible options for solution
  - Understand pros and cons of each solution option
  - Design a networking related solution for implementation

What would the world be like **without** computer networks?



Networks are very complex !

Move the bits,

- In large quantities
- Everywhere
- Cheaply
- As fast as possible
- Securely
- With quality of service

- Hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

- Future networks will be even more complex!

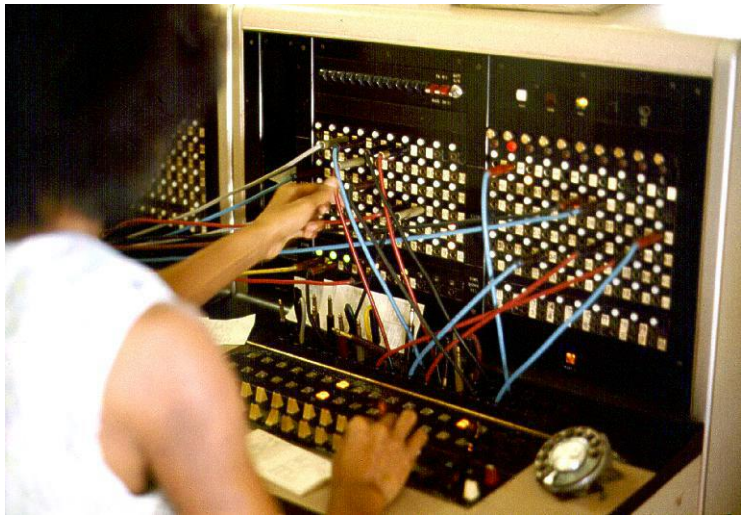


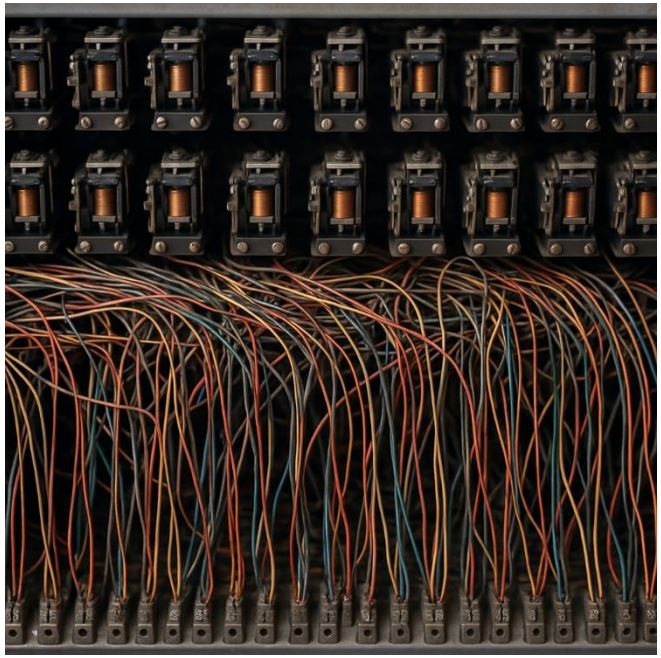
This course's challenge

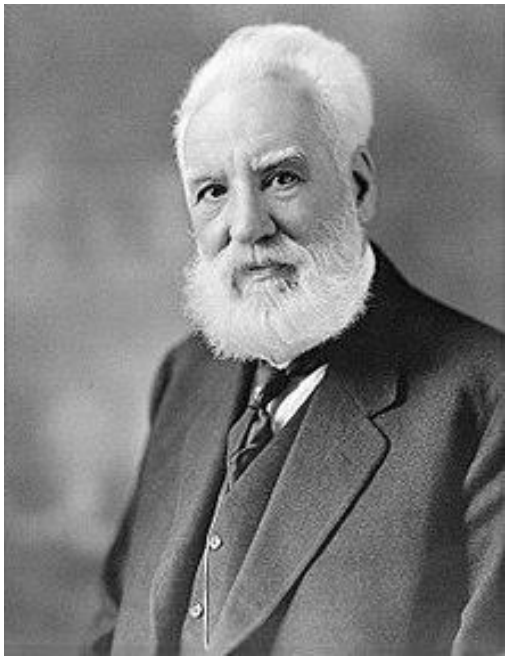
- To discuss this **complexity** in an **organized** way, that will make today's computer networks (and their limitations) more comprehensive.
- **Identification**, and understanding relationship of complex system's pieces.

Telephone (1876)





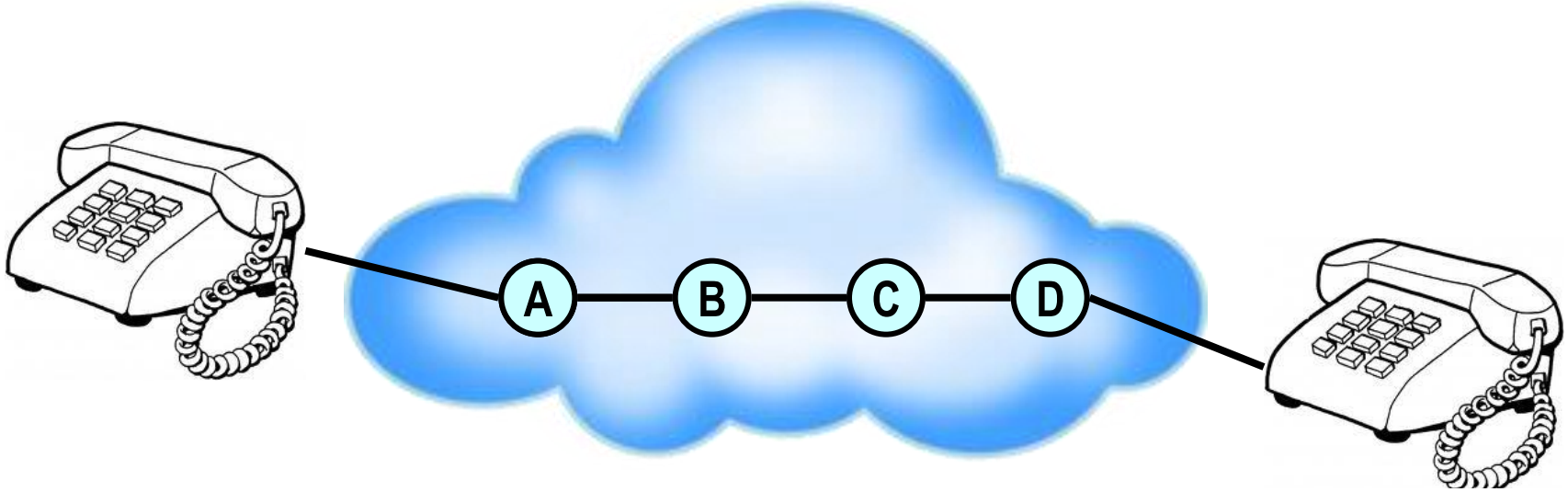




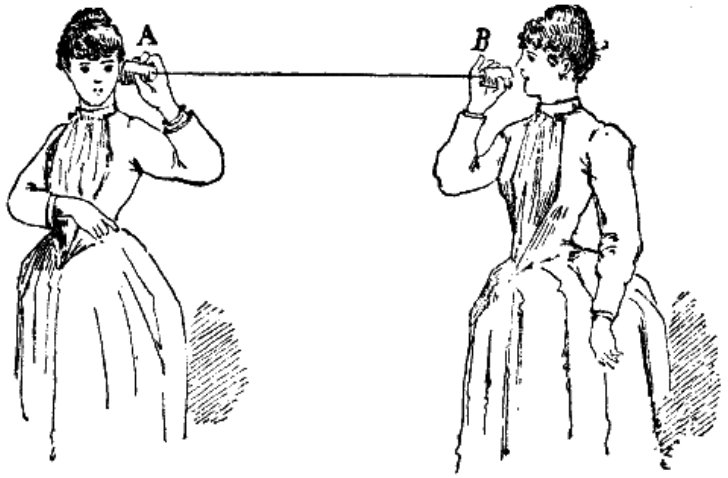
Alexander Graham Bell
(March 3, 1847 – August 2, 1922)

Was awarded the first U.S. patent for the invention of the telephone in 1876

Telephone = circuit switching
מיתוג מעגלים

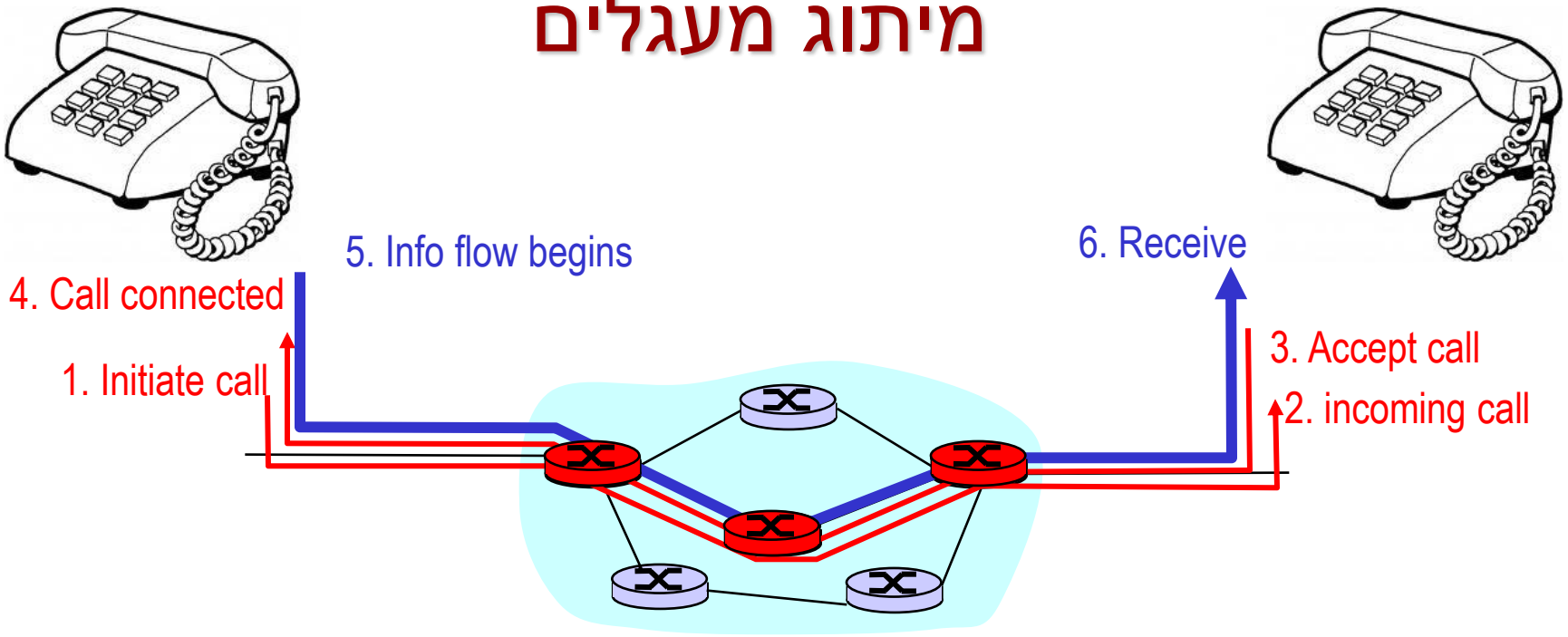


- Point-to-point links
- Directly connect the users wishing to communicate
- Use dedicated communication links



Telephone = circuit switching

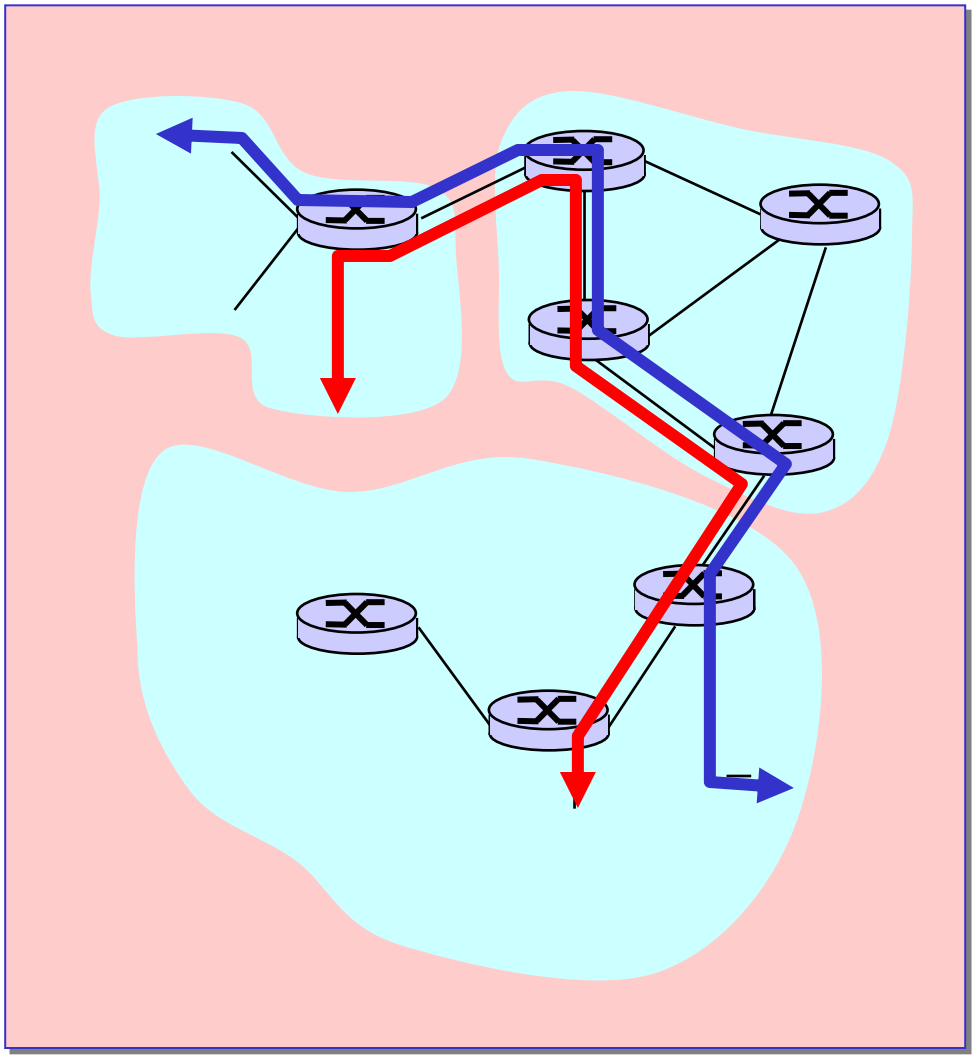
מיתוג מעגלים



Once the call is connected, the **route never changes** during the call.

Network core: circuit switching

End-end resources reserved for “call”



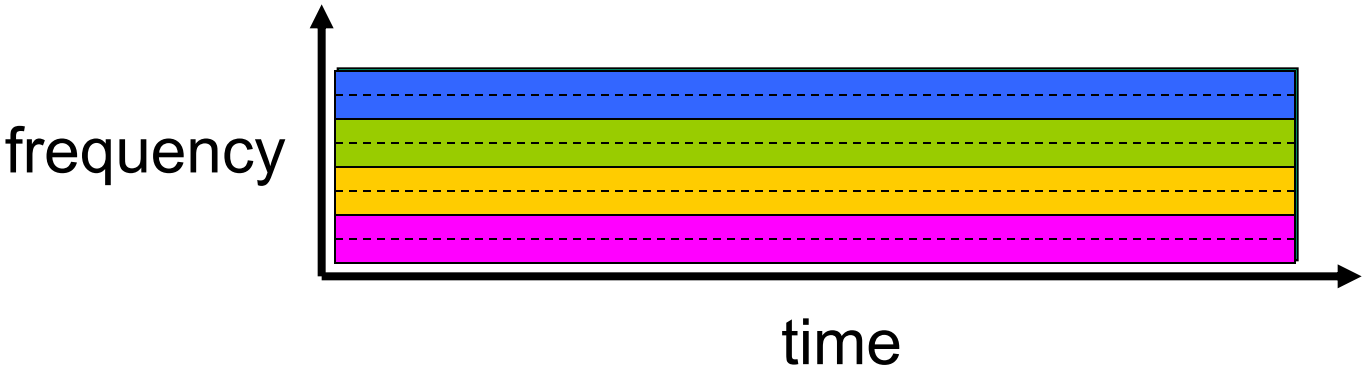
Circuit Switching: FDM and TDM

FDMA

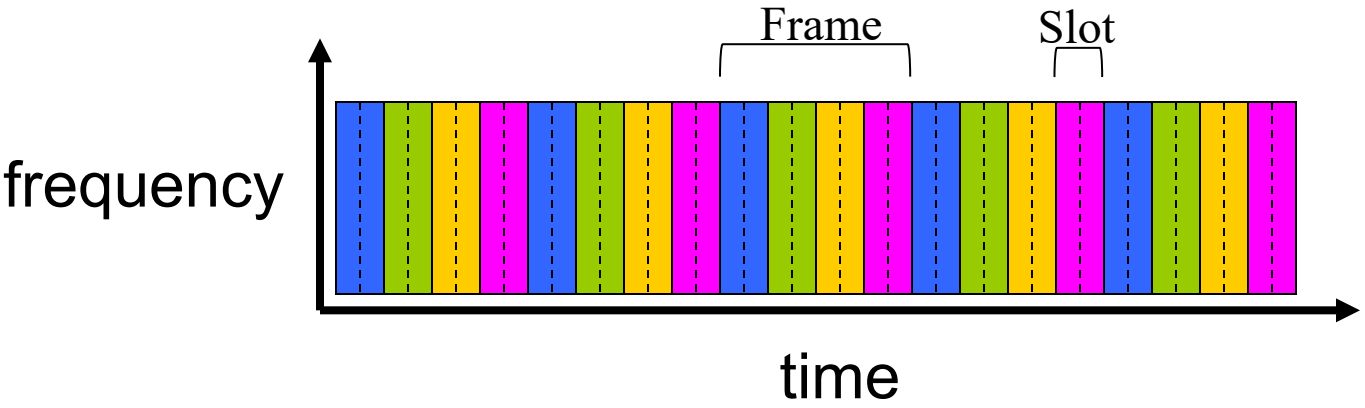
Example:

4 users





TDMA

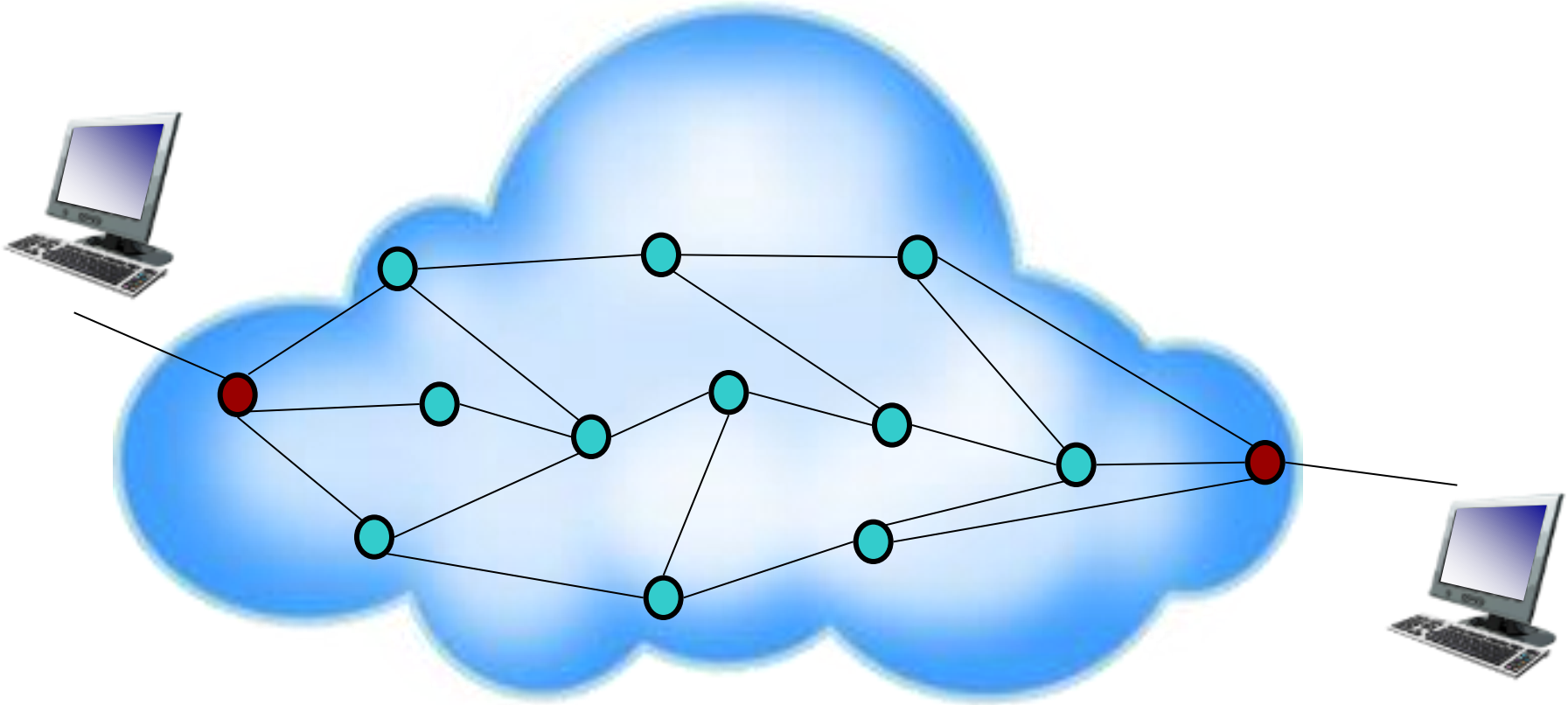


Question

- How long does it take to send a file of 7Mbits from host A to host B over a circuit-switched network?
  - All links are 3.5 Mbps
  - Each link uses **TDM** with 10 slots/sec
    (that is, each frame includes 10 slots)
  - It takes 750 msec for call setup

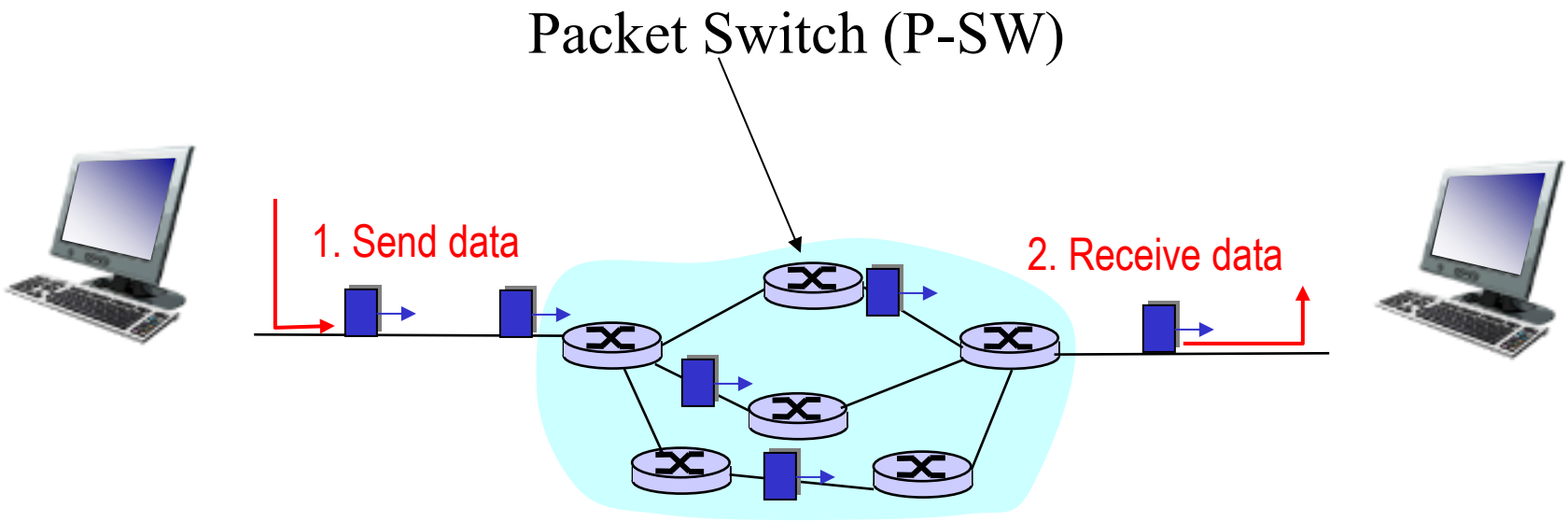


Data networks = packet switching
מיתוג מנות



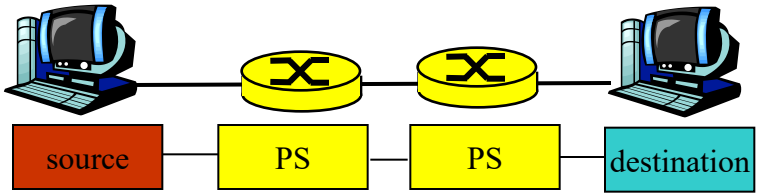
- Set of interconnected nodes exchange information.
- Many links allow more than one path between every 2 nodes.

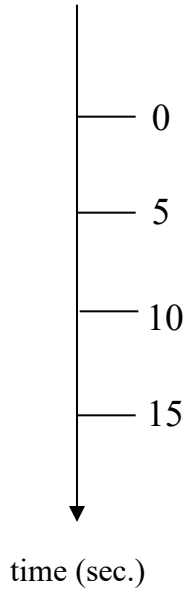
Data networks = packet switching
מיתוג מנות



Assumption for now (Store and Forward):
Packets cannot be forwarded until they are received in full by the switch

Packet switching





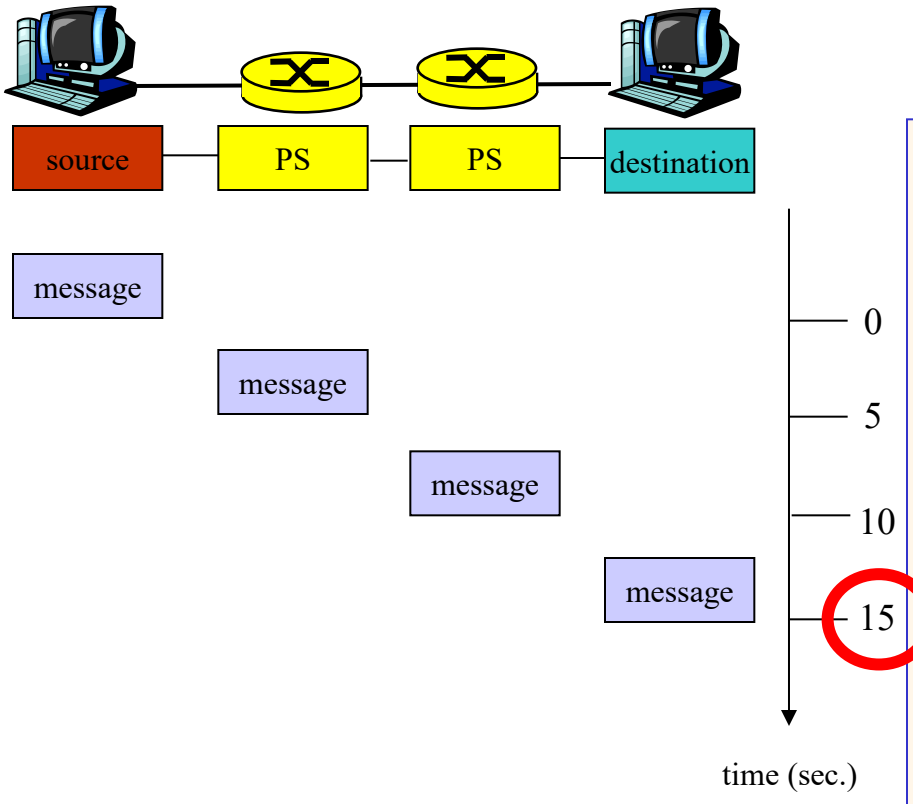
store and forward behavior

link transmission rate 1.5 Mbps

message 7.5 Mbit

packets 1

Packet switching



store and forward behavior

link transmission rate 1.5 Mbps

message 7.5 Mbit

packets 1

packets size 7.5Mbit

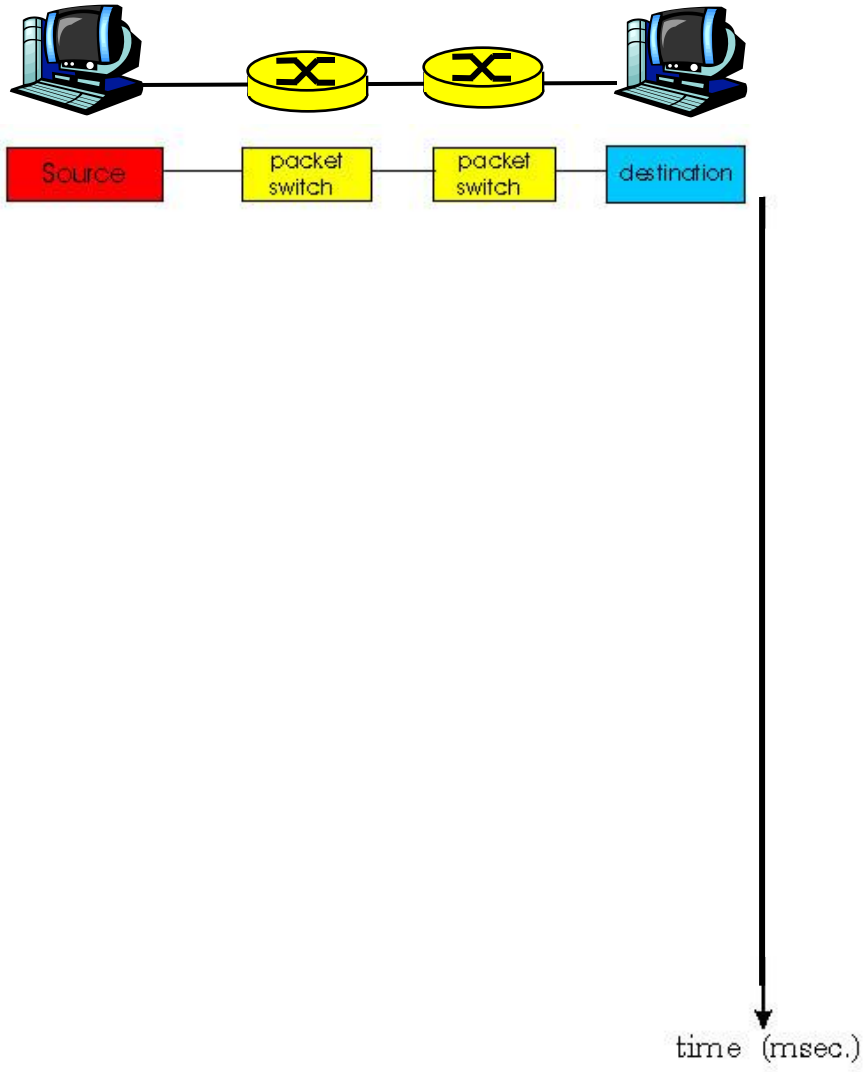
$$\frac{7.5_{Mbit}}{1.5_{Mbps}} = 5_{sec}$$

to move a packet/message

5000 packets how much time?

Packet switching

pipelining: each link works in parallel



store and forward behavior

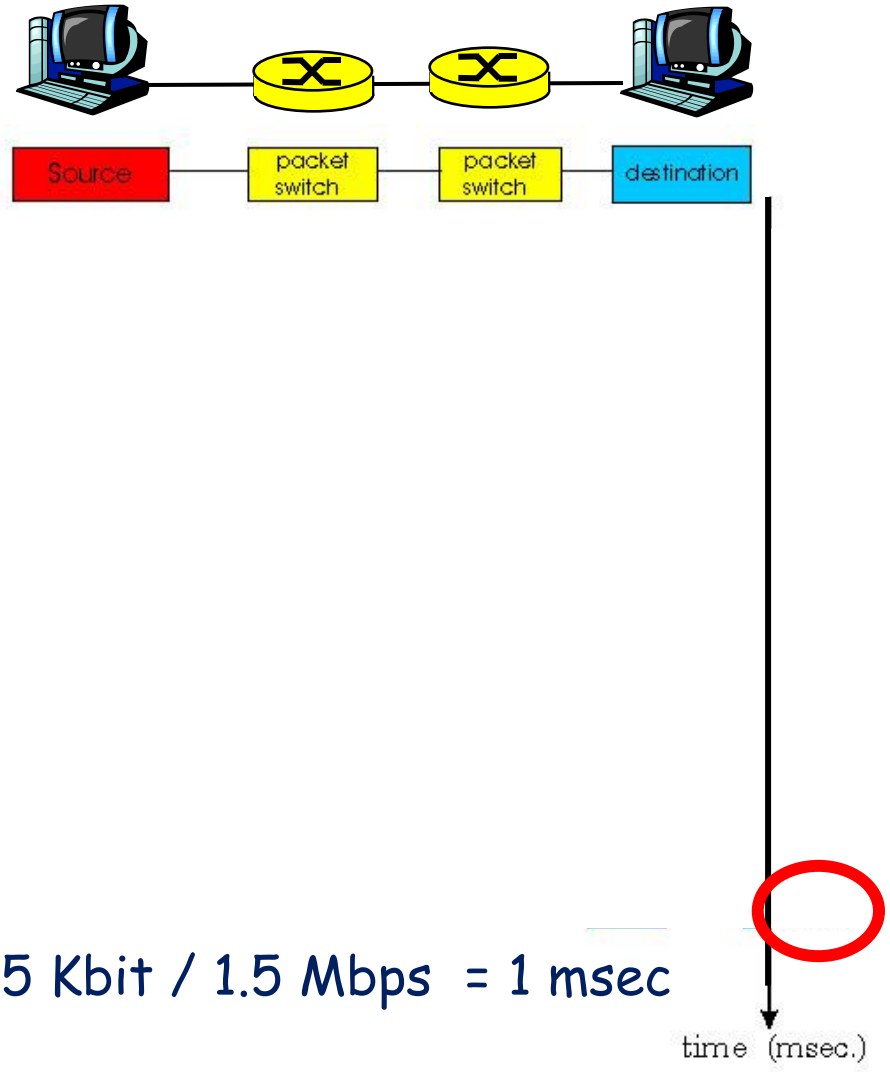
link transmission rate 1.5 Mbps

message 7.5 Mbit

packets 5000

Packet switching

pipelining: each link works in parallel



1.5 Kbit / 1.5 Mbps = 1 msec

store and forward behavior

link transmission rate 1.5 Mbps

message 7.5 Mbit

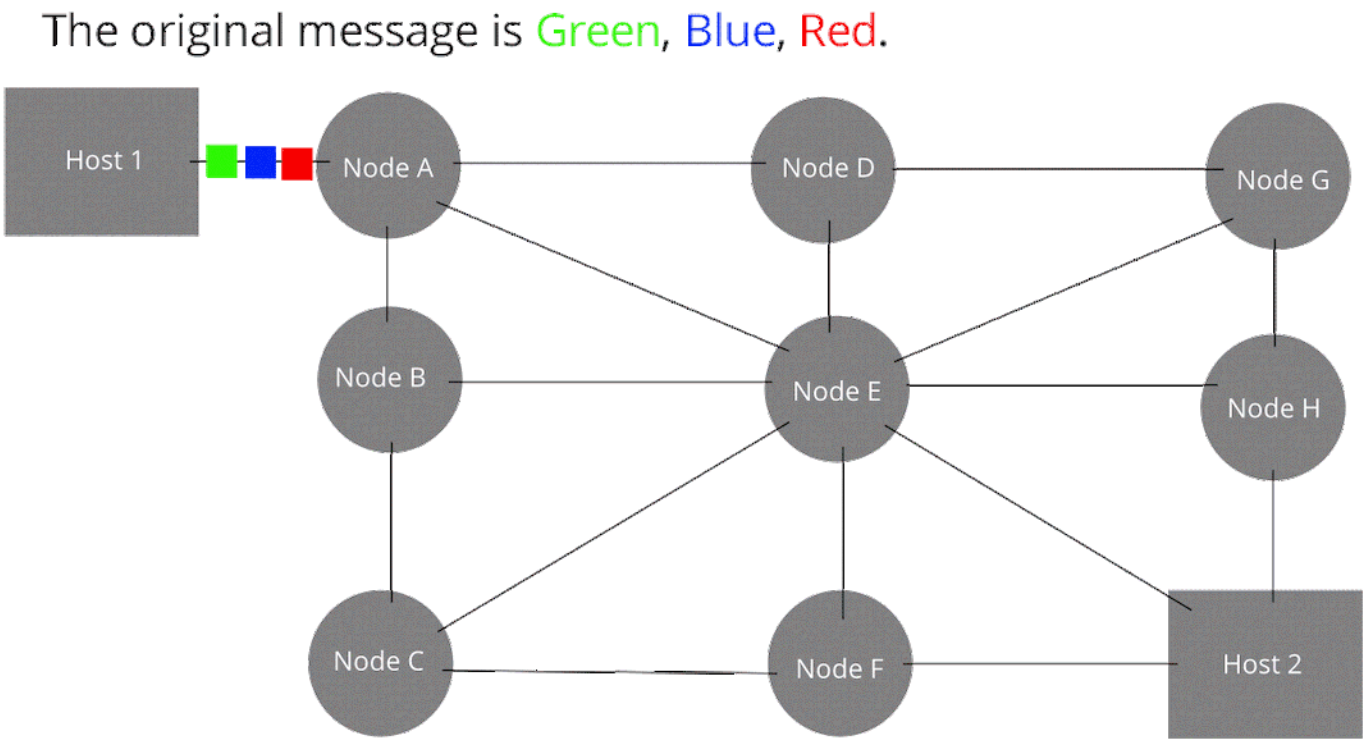
packets 5000

packets size 1.5Kbit

$$\frac{1.5_{kb}}{1.5_{Mbps}} = 1_{msec}$$

to move a single packet

Packet switching



Networking issues

Telephone

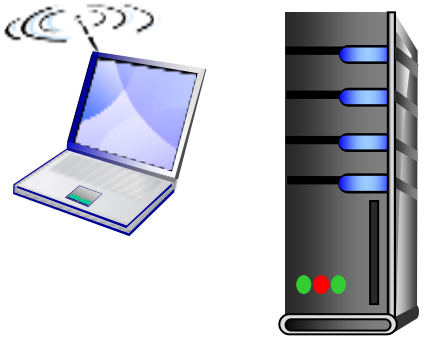


- Addressing
- Routing
- Information Units

Internet

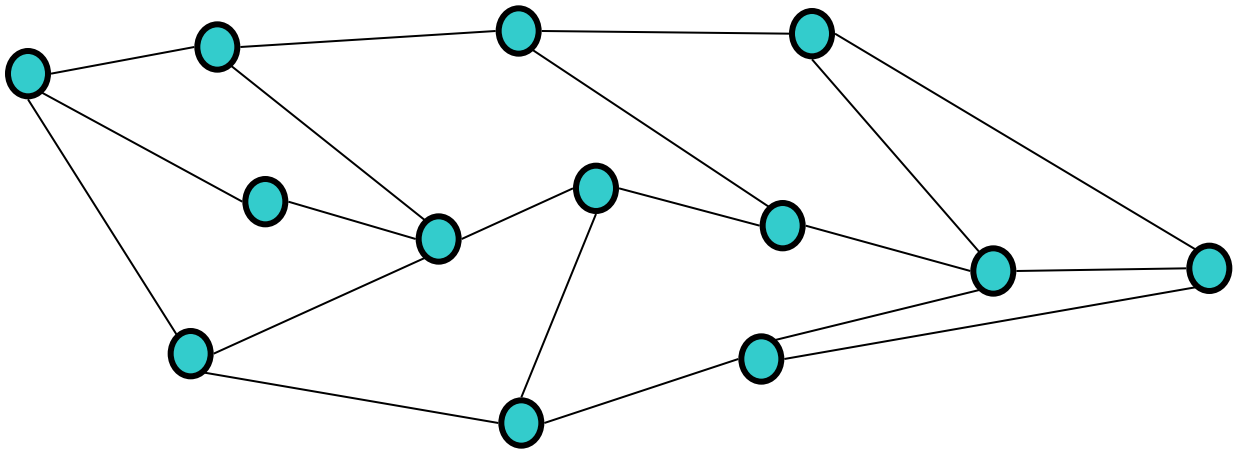




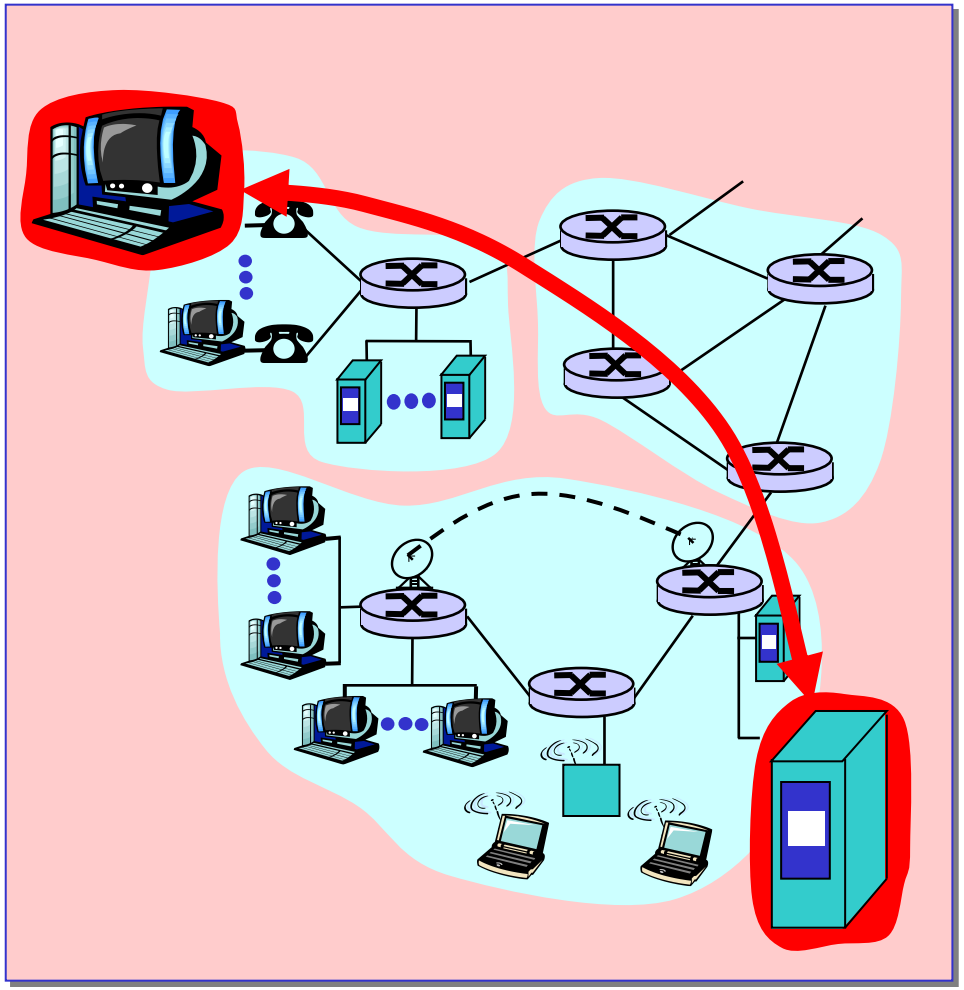


Terminology

- Circuit switch
- Packet switch = router = נתב

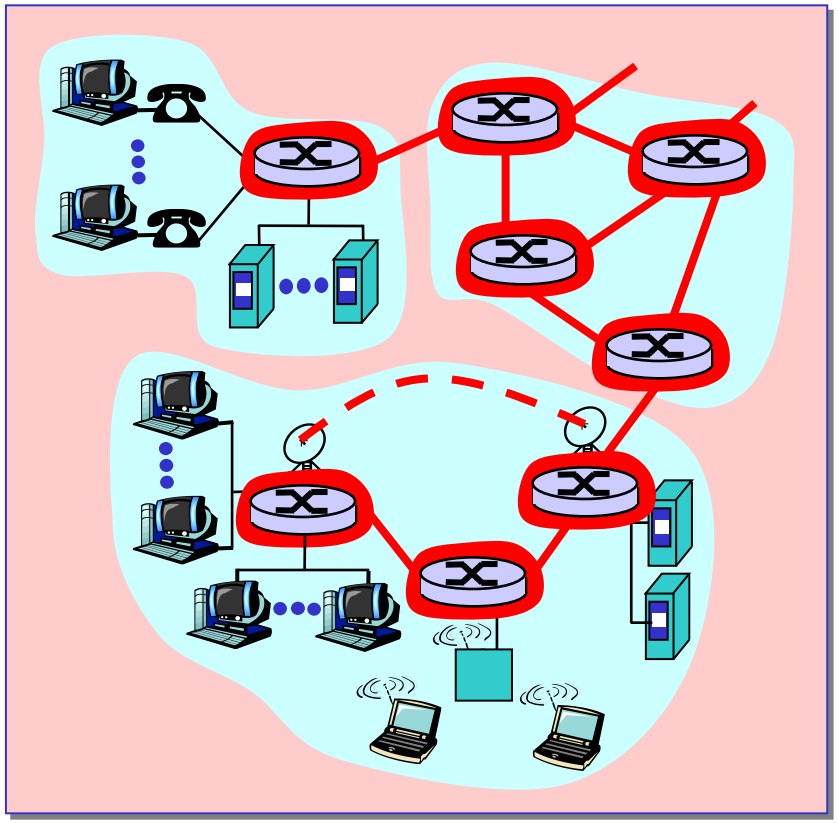


The network edge



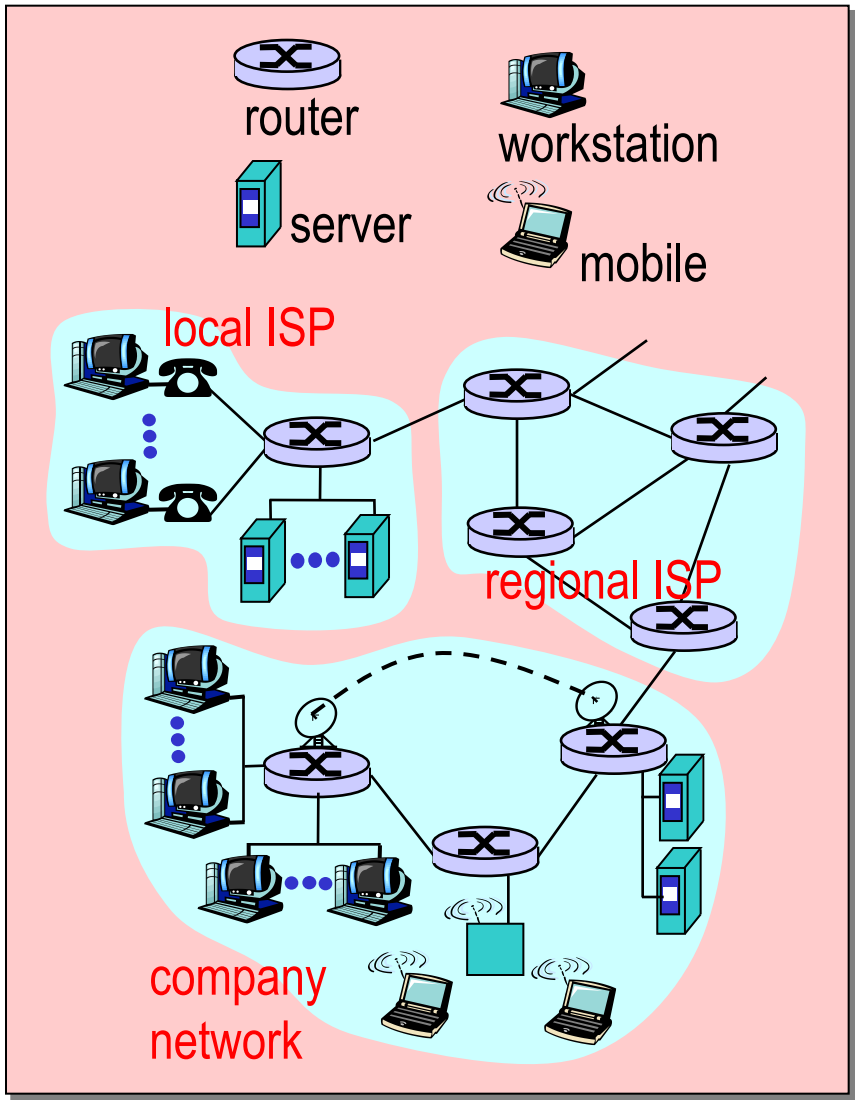
The network core

How data is transferred through the net?



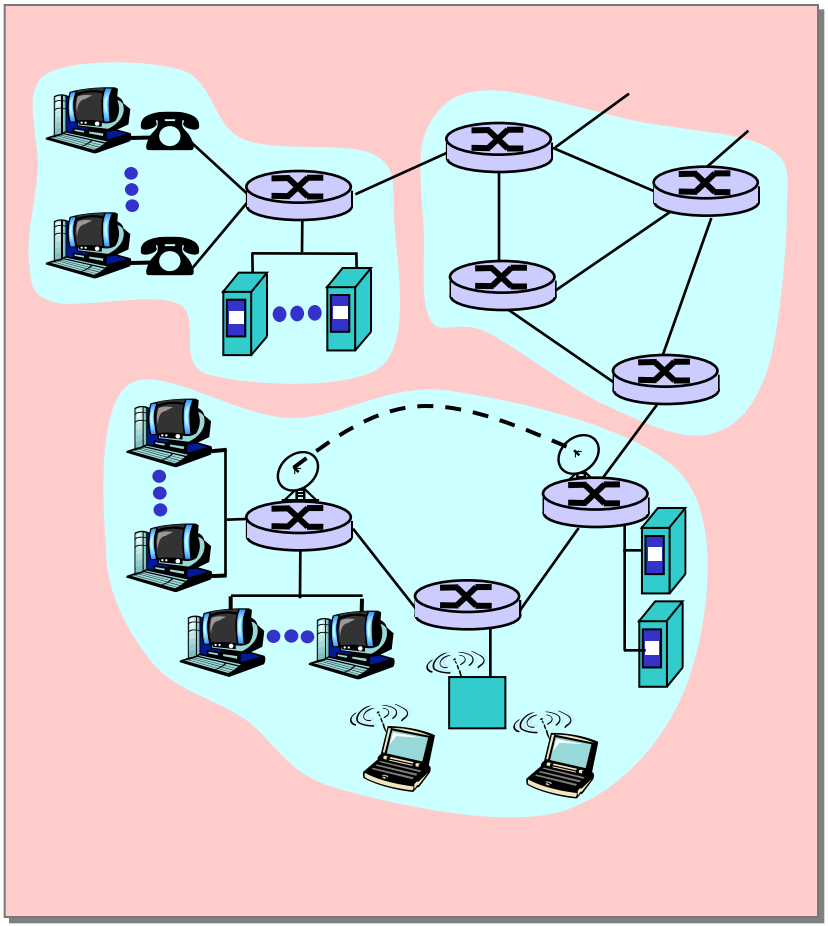
What's the Internet: "nuts and bolts" view

- *Host (end systems)*
  - millions of connected computing devices running network apps
- *Communication links*
  - fiber, copper, radio, satellite
- *Routers*
  - forward packets (chunks) of data thru network
- *protocols*
  - control sending, receiving of msgs
  - e.g., TCP, UDP, IP, HTTP, FTP, OSPF, RIP, PPP, Ethernet …
- *Internet:* "network of networks"

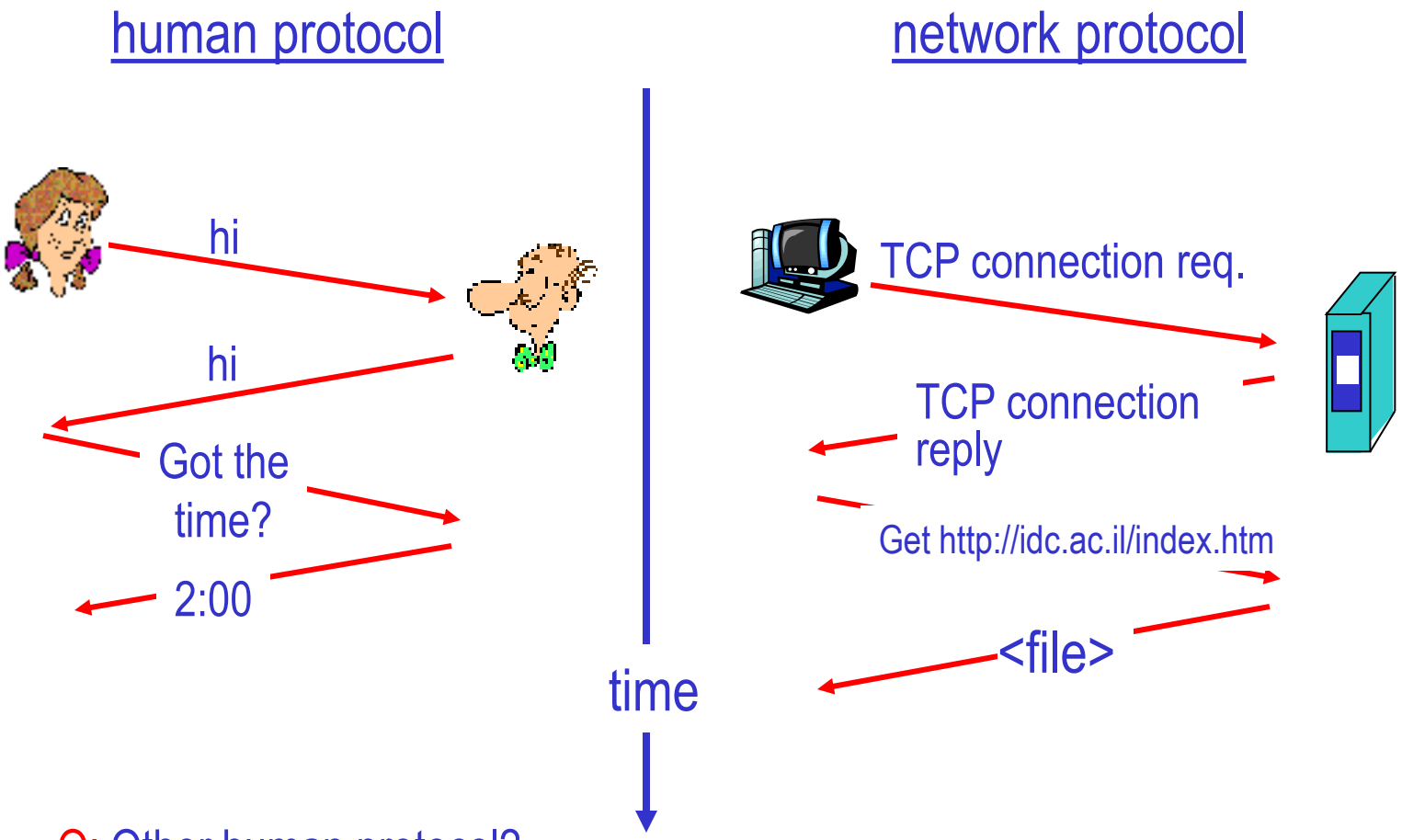


What's the Internet: a service view

- communication *infrastructure* enables distributed applications
  - WWW, email, games, e-commerce, database., voting,
  - more?
- communication services provided
  - connectionless
  - connection-oriented



What's a protocol?

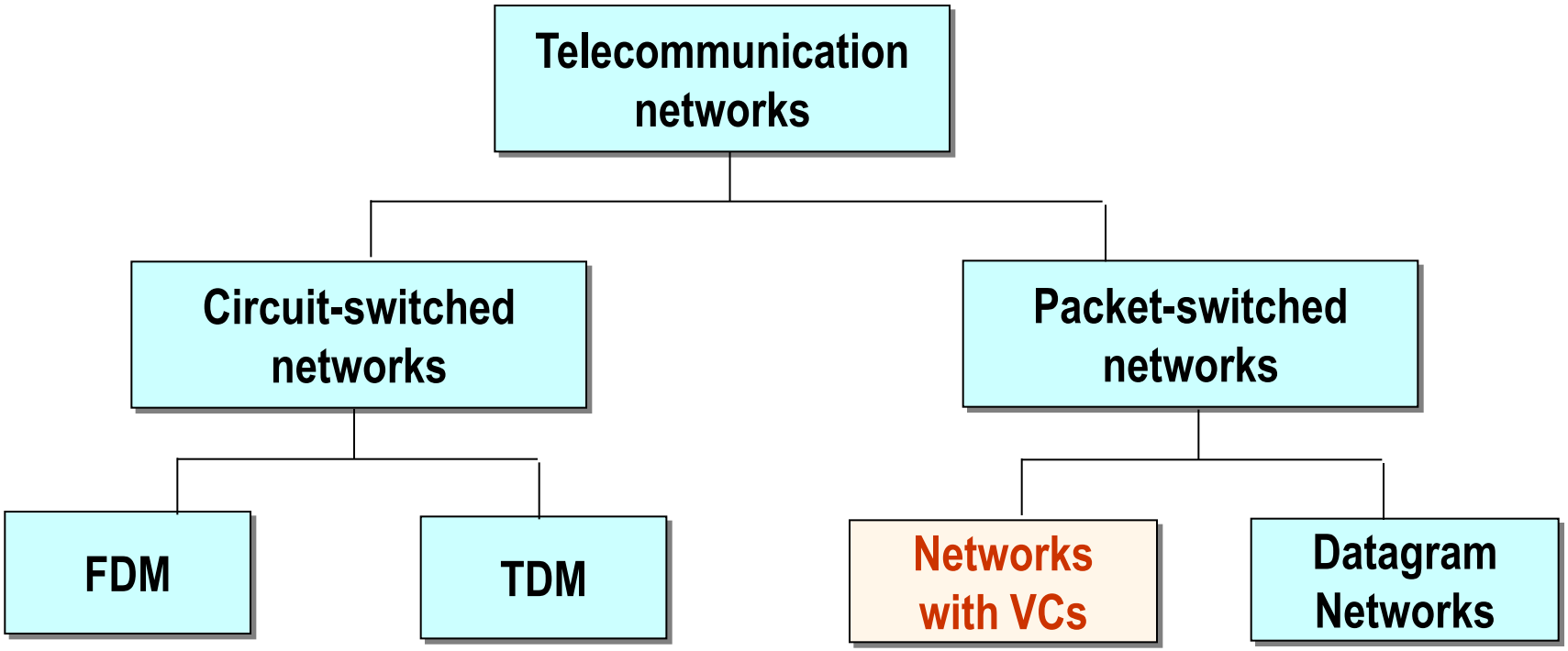


What's a protocol?

Protocols define:

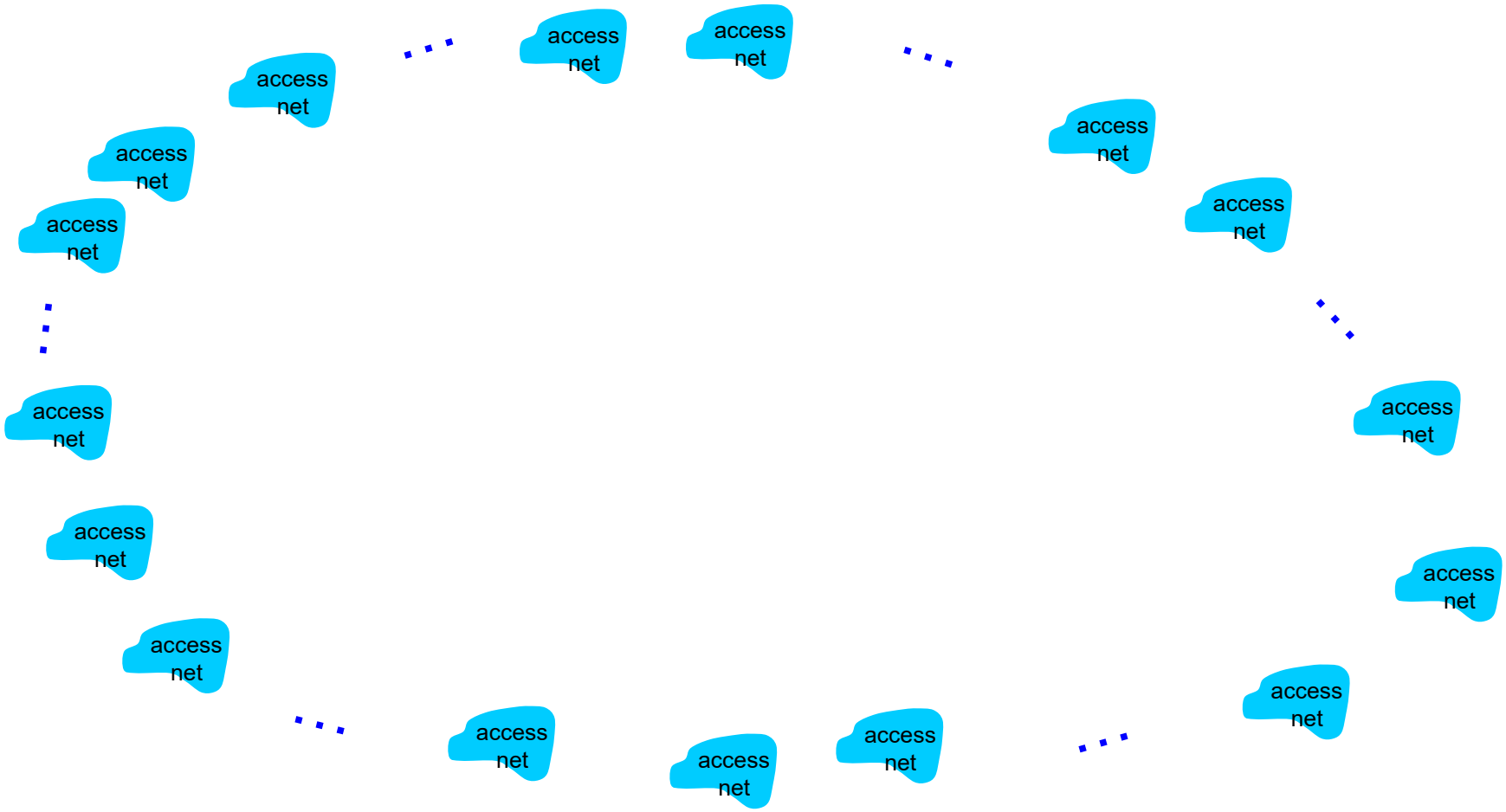
- Format
- Order of msgs sent and received among network entities,
- Actions taken on msg transmission, receipt

Network Taxonomy



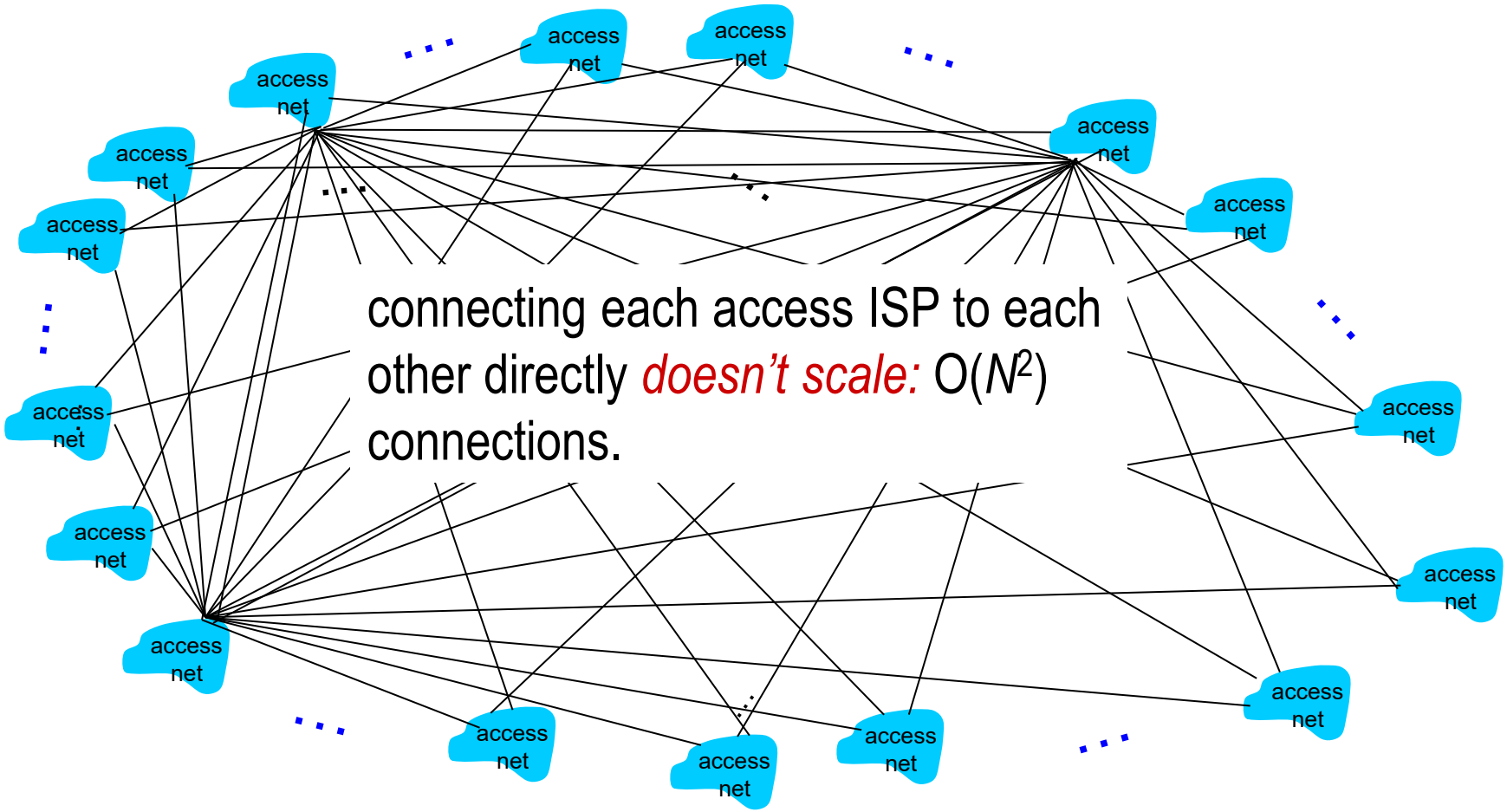
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



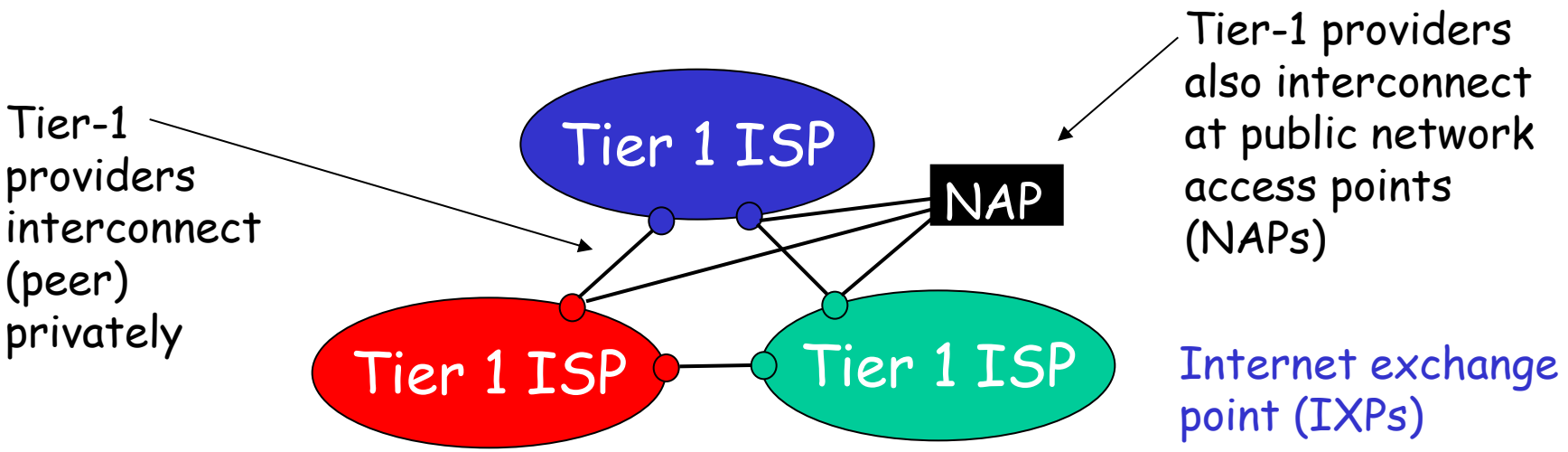
Internet structure: network of networks

Option: connect each access ISP to every other access ISP?



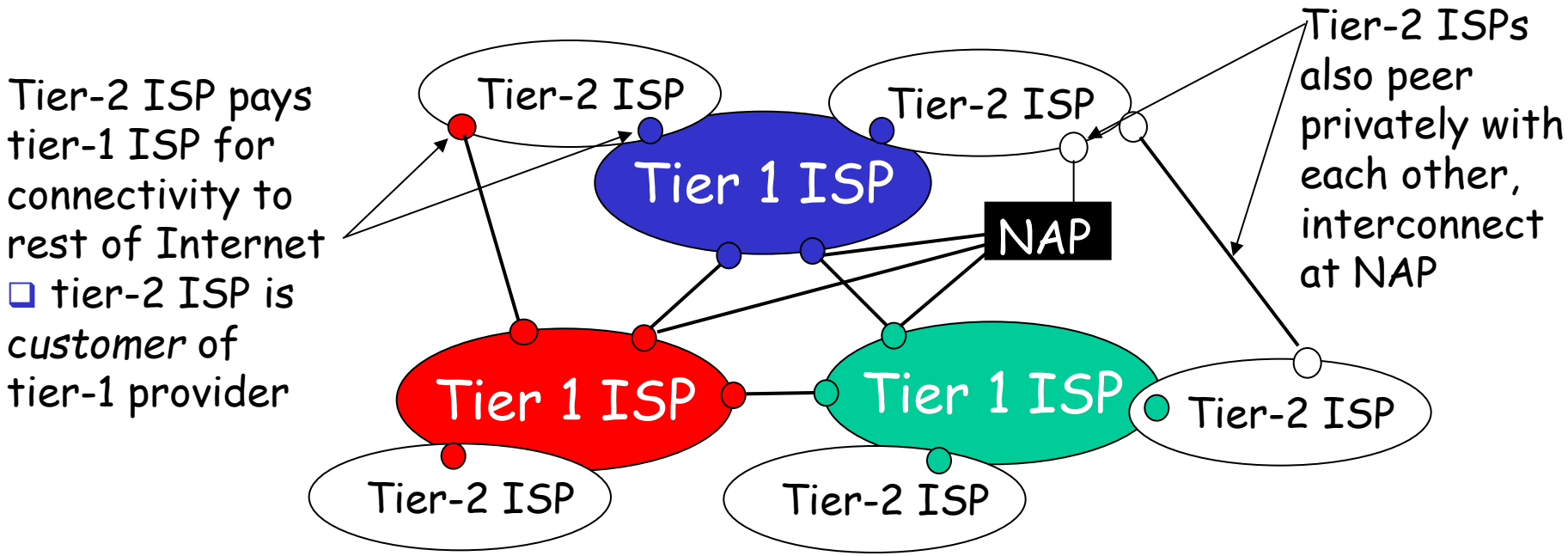
Internet structure: network of networks

- roughly hierarchical
- at the center: "Tier-1" (Internet Service Providers) ISPs - (national/international backbone providers) (e.g., UUNet, BBN/Genuity, Sprint, AT&T)



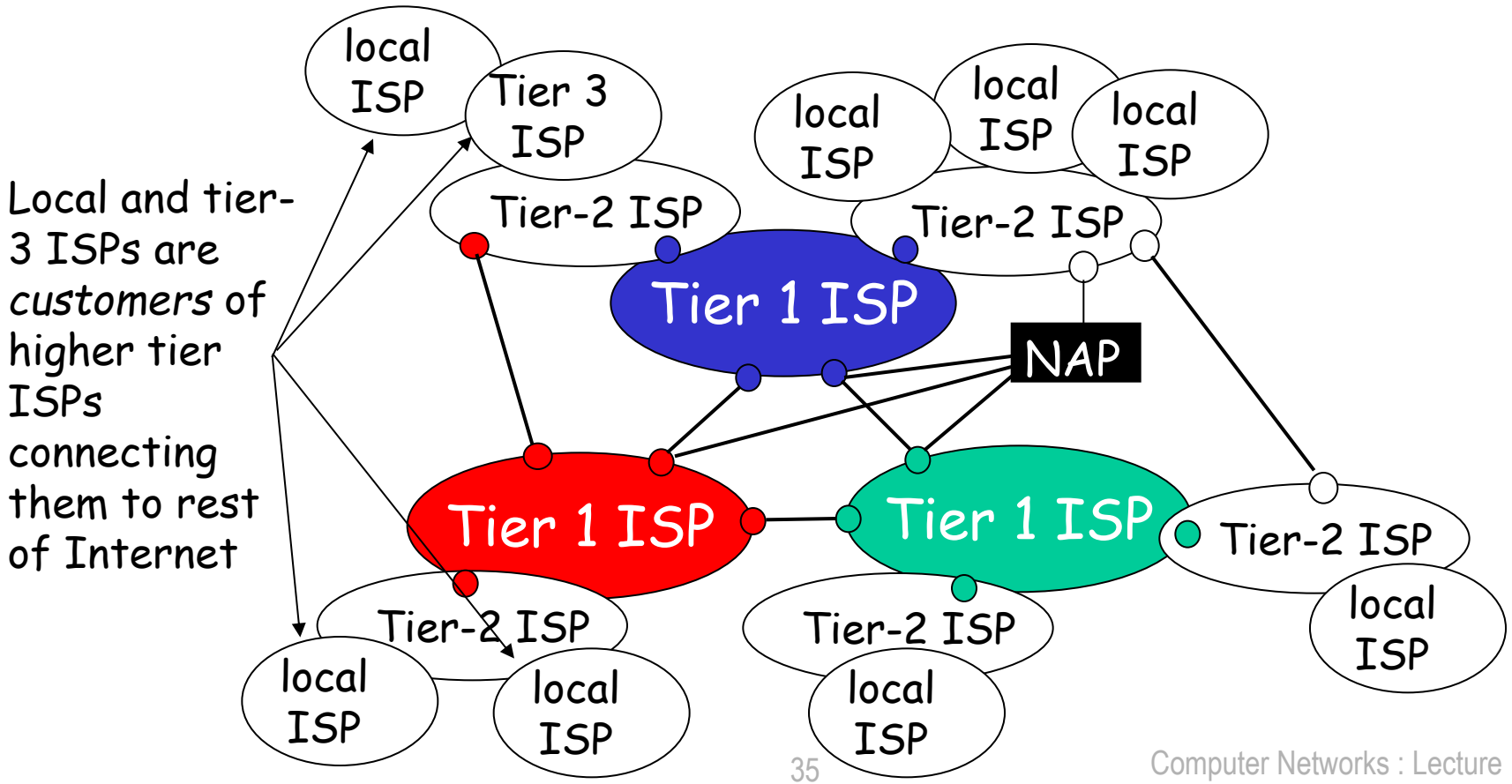
Internet structure: network of networks

- "Tier-2" ISPs: smaller (often regional) ISPs
  - Connect to one or more tier-1 ISPs, possibly other tier-2 ISPs



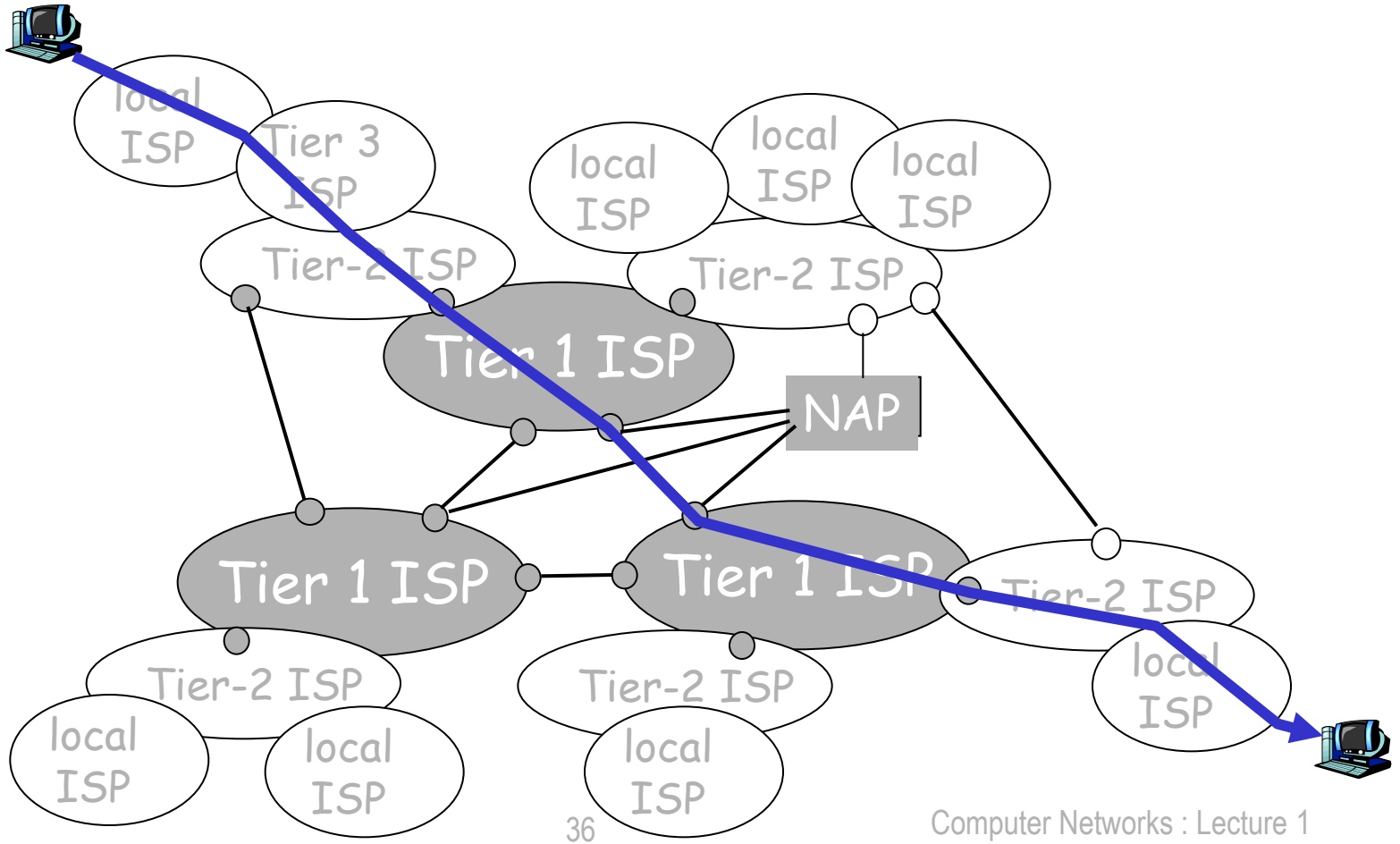
Internet structure: network of networks

- "Tier-3" ISPs and local ISPs
  - last hop ("access") network (closest to end systems)

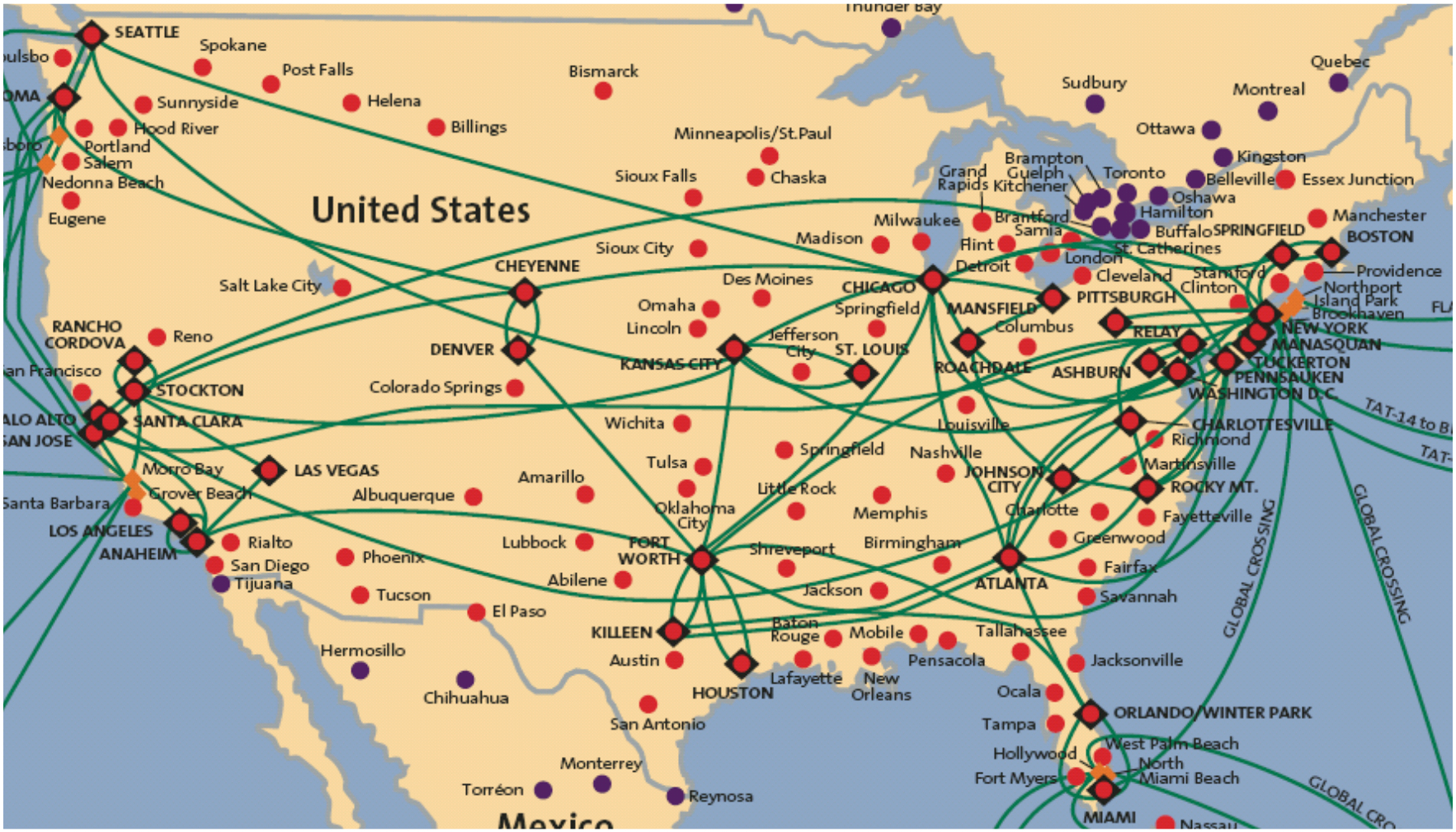


Internet structure: network of networks

- a packet passes through many networks!



Tier-1 ISP: e.g., Sprint



Protocol Layers

Networks are complex!

- many "pieces":
  - hosts
  - routers
  - links of various media
  - applications
  - protocols
  - hardware, software

Question

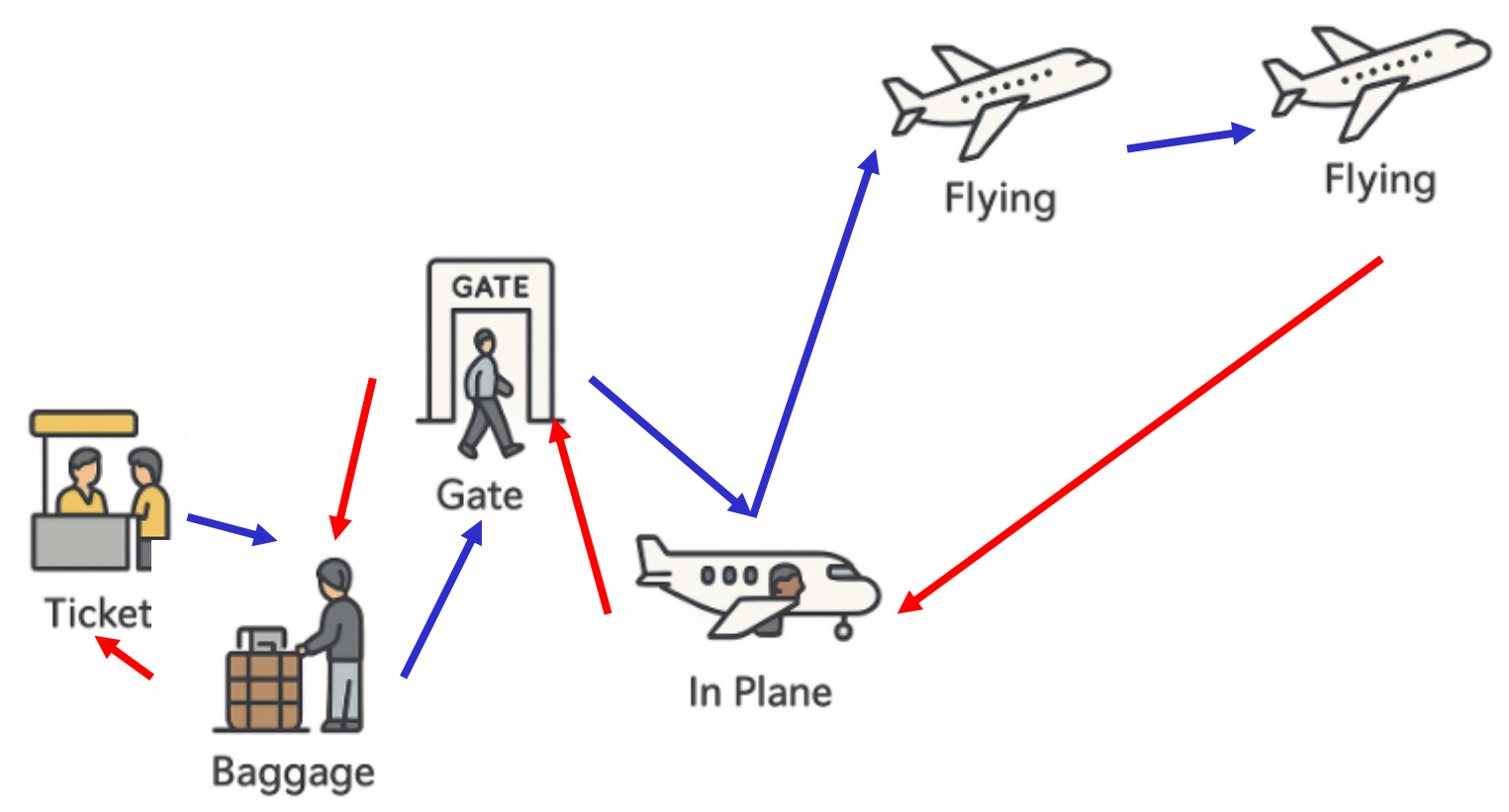
Is there any hope of *organizing* structure of network?

Or at least our discussion of networks?

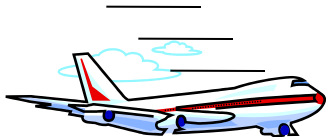
Answer: layers

A way for *organizing* structure of network

Describing a person flying
Use location-based thinking



Describing a person flying
Layer based thinking

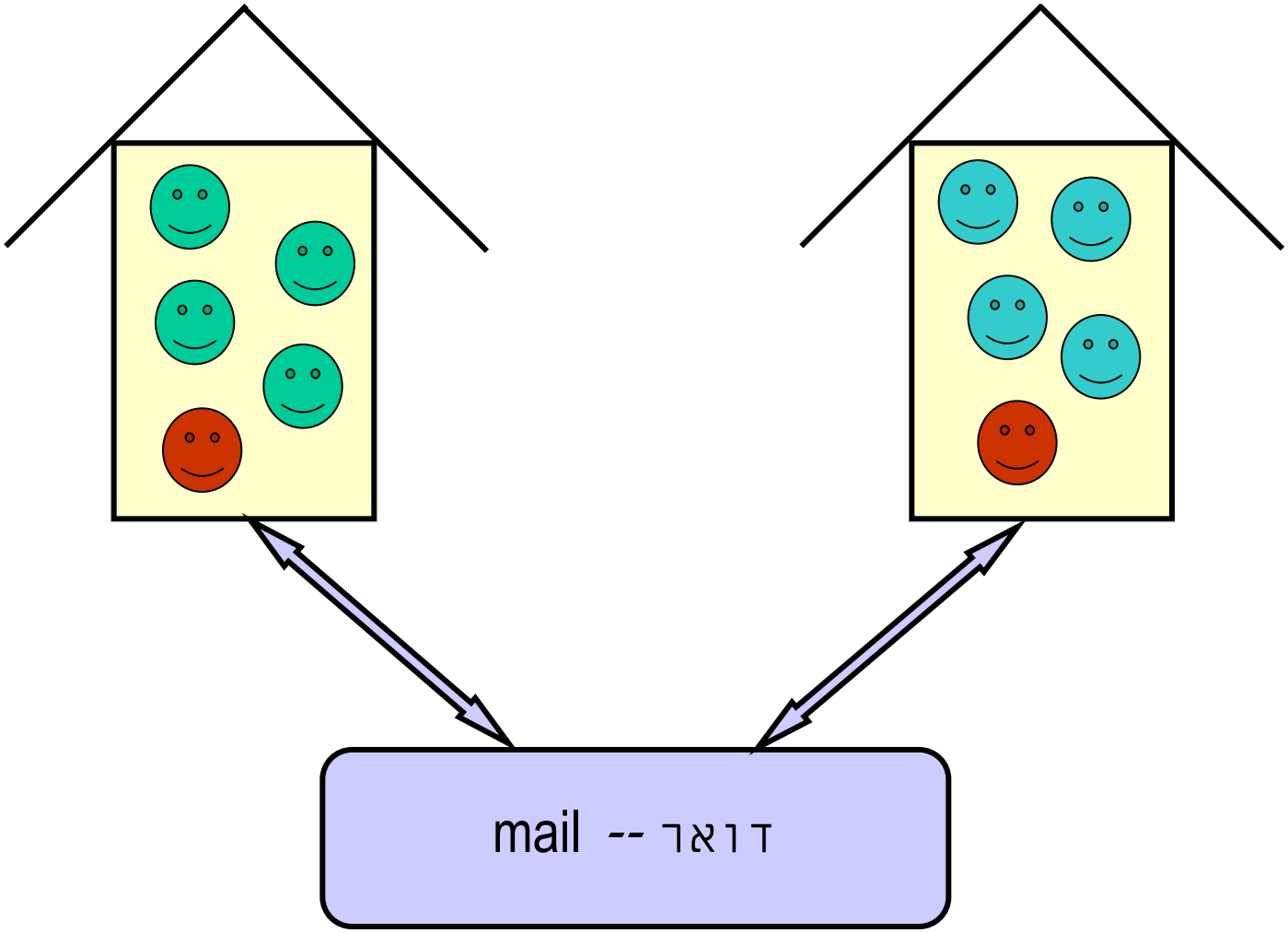


departure airport		intermediate air-traffic control centers		arrival airport	
ticket (purchase)				ticket (complain)	ticket
baggage (check)				baggage (claim	baggage
gates (load)				gates (unload)	gate
runway (takeoff)				runway (land)	takeoff/landing
airplane routing		airplane routing	airplane routing	airplane routing	airplane routing

Pros and Cons of layering

Pros	Cons
Easy breaking and understanding of complex systems – solving the big problem via a list of smaller ones	Performance overhead (extra headers, copies, context switches)
Modularity & component reuse	Duplicate functionality across layers (e.g., error control)
Parallel development; independent evolution of layers – allows	Harder cross-layer optimization; protocol ossification
Easier testing & troubleshooting (isolate by layer)	Added latency from per-layer processing
Scalability and flexible deployment	End-to-end debugging across layers is complex

One more example …



Internet protocol stack

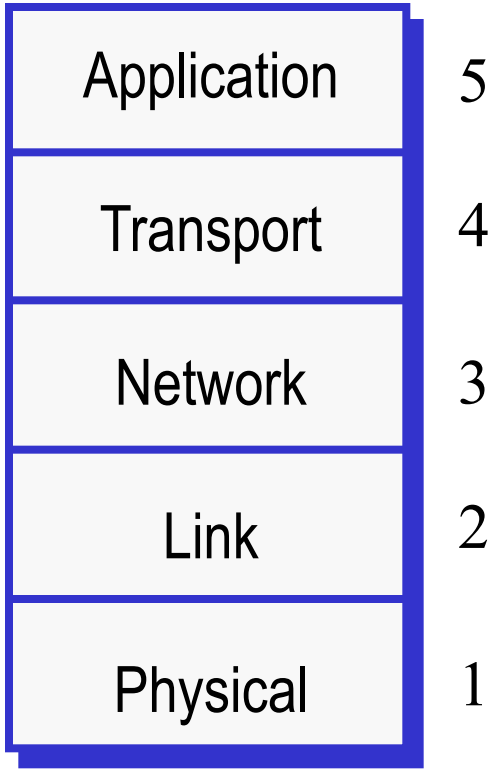
5 ❑ application: supporting network applications
- HTTP, DNS, SMTP, POP3, IMAP, SSL, TLS

4 ❑ transport: application to app. data transfer
- TCP, UDP

3 ❑ network: routing of datagrams from source to destination (host to host)
- IP, routing protocols

2 ❑ link: data transfer between neighboring network elements
- Ethernet, 802.111 (WiFi), PPP

1 ❑ physical: bits “on the wire”



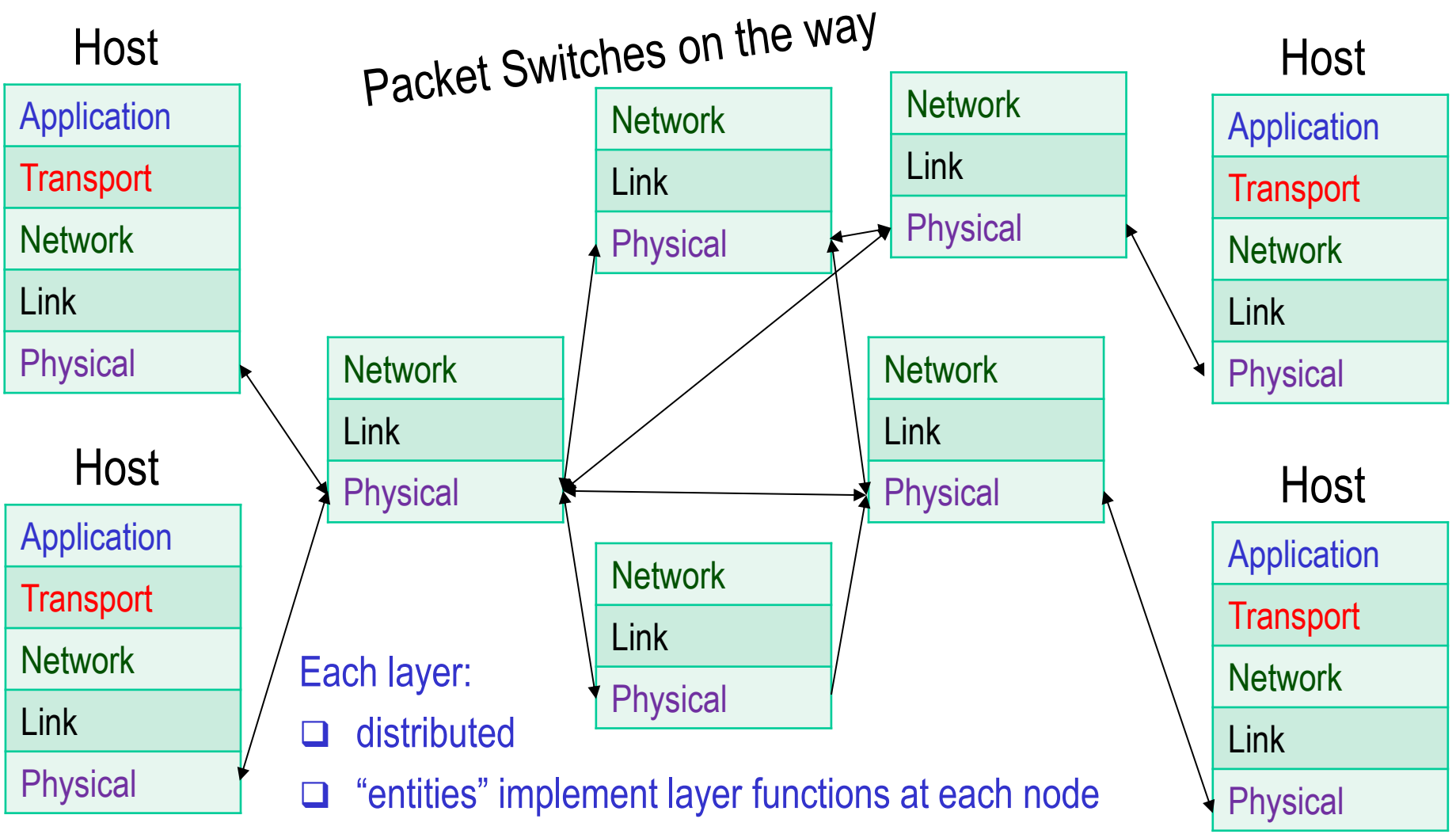
ISO/OSI reference model

- *presentation:* allow applications to interpret meaning of data, e.g., encryption, compression, …
- *session:* synchronization, checkpointing, recovery of data exchange
- Internet stack "missing" these layers!
  - these services, *if needed,* must be implemented in the application



Layering: communication

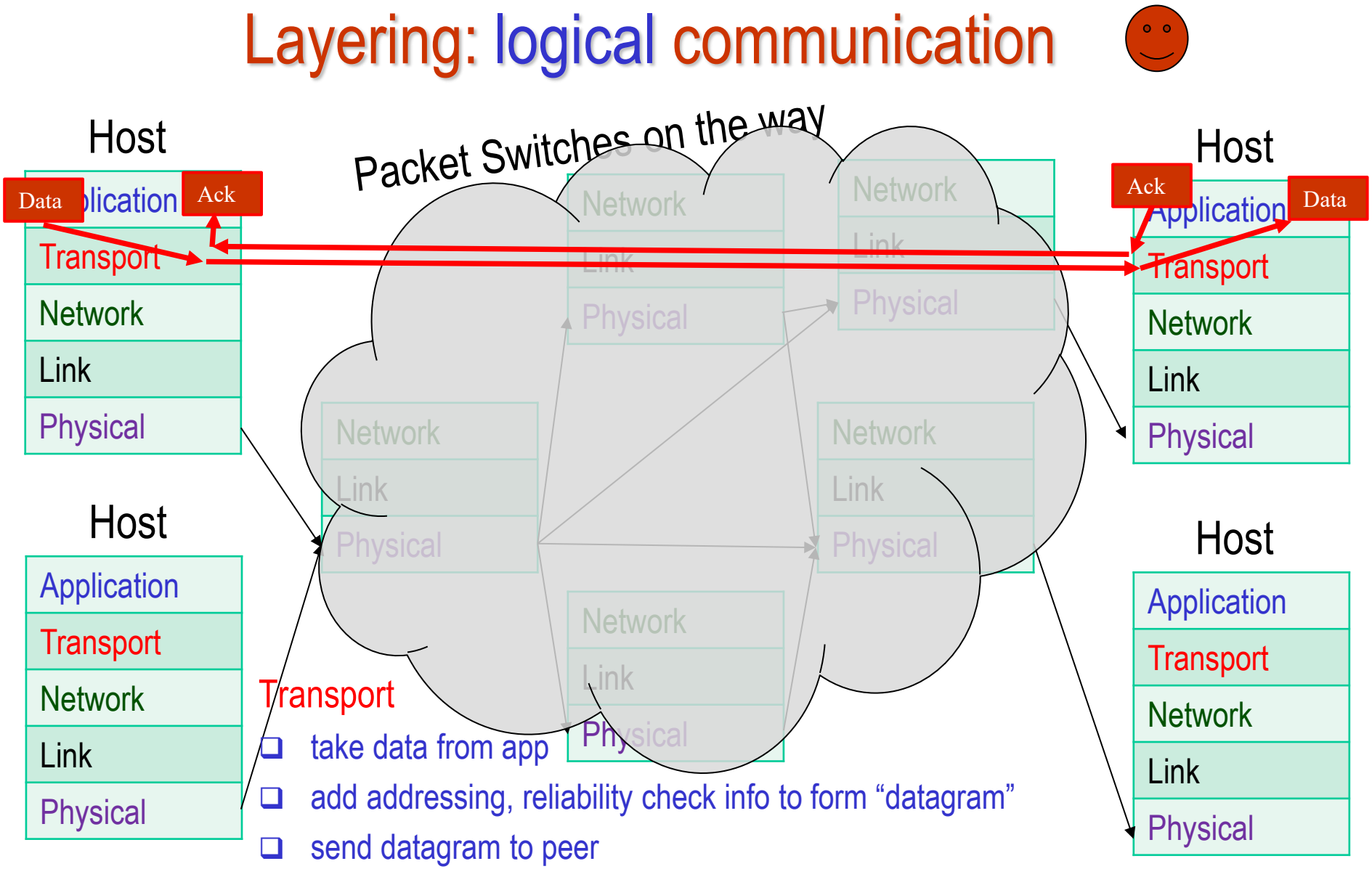




Each layer:

- distributed
- "entities" implement layer functions at each node
- "entities" perform actions, exchange messages with peers
- Components implement what is needed for them

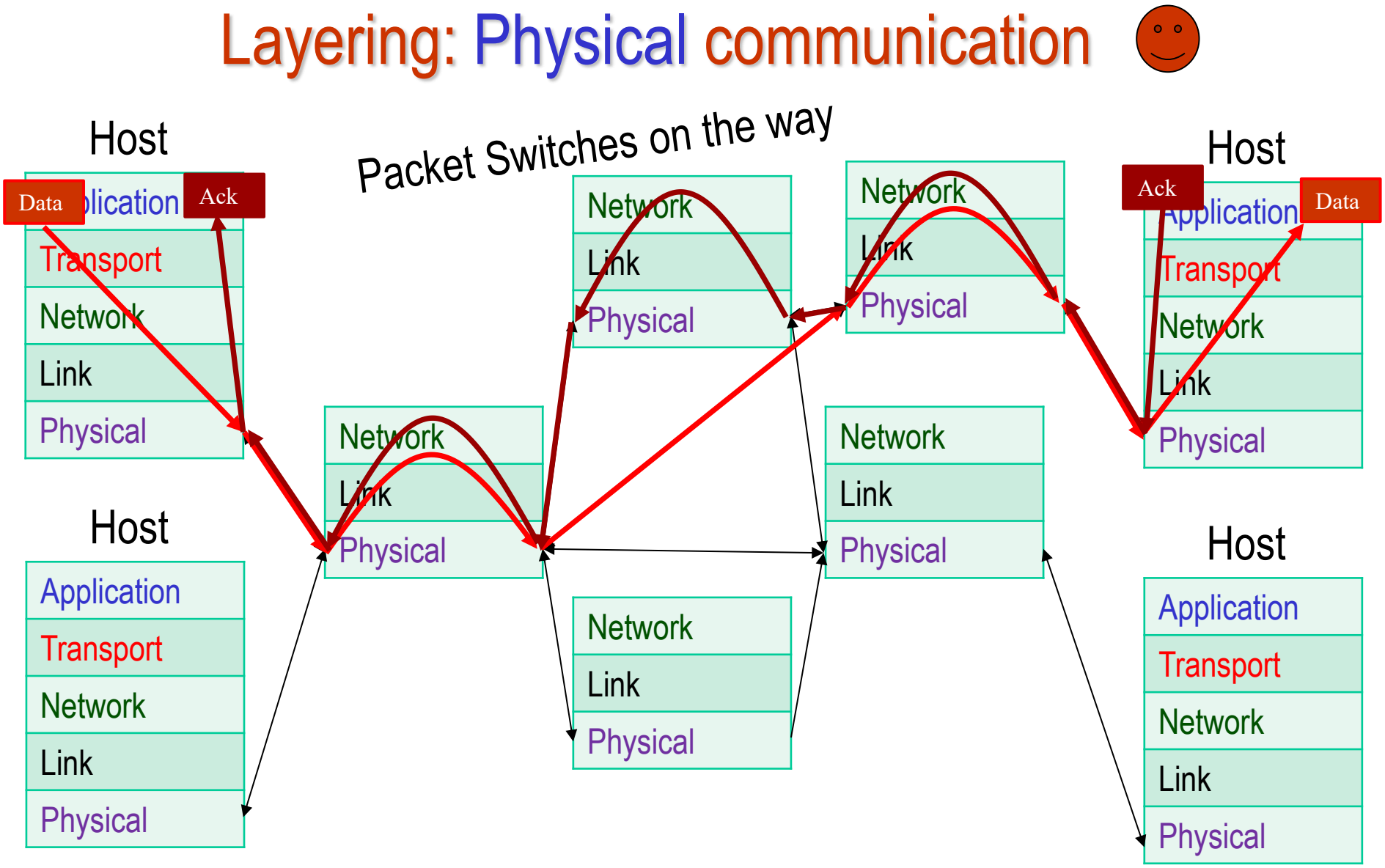
Layering: logical communication



Transport

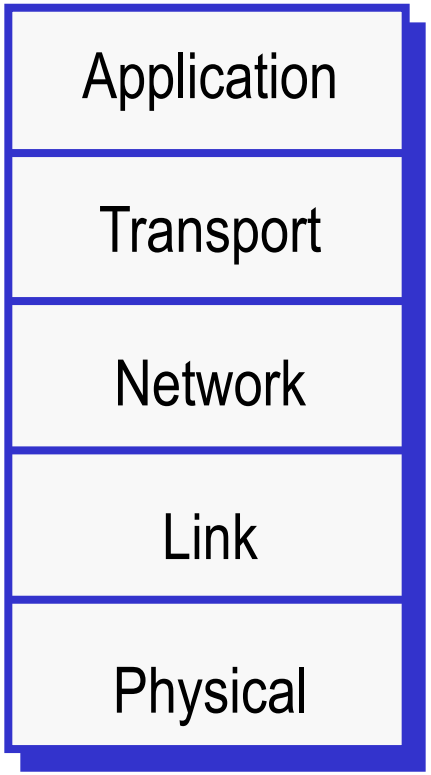
- take data from app
- add addressing, reliability check info to form "datagram"
- send datagram to peer
- wait for peer to ack receipt (sometimes)
- analogy: post office

Layering: Physical communication



Course plan: 13 weeks

- Introduction — 2



2-3

2-3

3-4

1-2

- Distributed Algorithms — 2

TRUE/FALSE Quiz

---

1 In a circuit-switched network, a circuit switch maintains (state) information for each connection passing through the switch.

- FALSE
- TRUE

---

2 Before sending a packet into a datagram network, the source must determine all of the links that the packet will traverse between source and destination.

- FALSE
- TRUE

---

3 Layers four and five of the Internet protocol stack are implemented in the end systems but not in the routers in the network core.

- FALSE
- TRUE

---

4 The Internet provides its applications two types of services, a TDM service and a FDM service.

- FALSE
- TRUE

---

5 A packet-switched network can guarantee a certain amount bandwidth between a sender (client) and the receiver (server).

- FALSE
- TRUE

---

6 When the flow of information in a network is connection-oriented, all the packets in the same connection are routed in the same route.

- FALSE
- TRUE

---

7 Usually circuit switching can support more users than packet switching.

- FALSE
- TRUE

The Internet: How it all started ?

The US Dept. of Defense decided to build a network that will:

- Connect all big computers in the US
- Continue to function after a nuclear attack
- With no central control

- Solution: computer network call ARPANET (1969)



Clip: The history of the Internet

- http://www.lonja.de/the-history-of-the-internet/
- http://www.youtube.com/watch?v=9hIQjrMHTv4&NR=1&feature=fvwp

